A Little Robustness Is All You Need: Leveraging Predictions for Contextual Optimization

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Abstract

A common approach for contextual optimization first trains a model to predict the unknown state from the context, and then takes the action that is optimal for the predicted state. We show that this follow-the-prediction policy can be brittle: for several standard problems, even a minimum-variance predictor with arbitrarily small RMSE can incur constant excess loss. We propose a simple policy that robustifies any black-box predictor by optimizing against the worst case in an ε-neighborhood of the prediction. We identify a problem-specific quantity, the *robust optimization gap*, that captures the price of this hedging, and prove a general bound on the loss. We instantiate the framework for contextual posted pricing, ski rental, and house flipping, deriving closed-form robust policies and $O(\eta^{2/3})$ loss bounds in all three settings, where η is the RMSE of the predictor. For posted pricing, this improves upon a result of Medina and Vassilvitskii [18], and with a substantially simpler argument.

1 Introduction

Contextual decision problems arise across computer science, operations research and economics, in settings such as personalized pricing, online advertising, and recommender systems. In these problems, a decision-maker observes a context x, commits to an action, and then receives a payoff determined by the action and a realized state θ — for example, a customer's willingness to pay, a user's preferences, realized demand, or the highest competing bid in an auction. The challenge arises from the uncertainty in the realized state as the decision-maker can only observe the the context x, which may not give direct information about the relevant state θ.

A natural approach to addressing this challenge is to develop models which attempt to predict the underlying state θ from the context x. This is well motivated by the increasing capabilities of AI/ML models for solving a variety of prediction tasks, the hope being that a more accurate prediction model leads to improved performance on the contextual decision problem. This is epitomized by the *predict-then-optimize* pipeline: use a prediction model h to estimate θ from x, and at decision time choose the action that would be optimal if the true state were $h(x)$. We call this the *follow-the-prediction* (FtP) policy. This policy is appealing since it cleanly separates prediction from optimization, allowing the implementation of each step to be carried out separately. However, it has been observed that naïvely constructing the prediction model without taking into account the objective of the downstream decision problem can lead to substandard results [5, 12, 15], even when the prediction model may have high accuracy. This is especially the case when payoffs can change discontinuously across a threshold, as in posted pricing, so even a small prediction error can lead to the estimate landing on the wrong side of the threshold and incurring a large loss.

To address this issue, there is a growing literature on tailoring prediction models for downstream contextual decisions, including the seminal works of Kao et al. [15] on directed regression, Bertsimas

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and Kallus [5] on prescriptive analytics, and Elmachtoub and Grigas [12] on smart predict-then-optimize. While these works offer great insight for developing prediction models that can be utilized for contextual decision making, they lose the original appeal of the predict-then-optimize framework. Instead of training the prediction model independently, where it may be possible to leverage the most recent developments from AI/ML to solve this task, these approaches can significantly complicate the training process by requiring the use of particular model structures or non-standard loss functions. One might instead hope that we can treat the prediction model as a black-box and post-process its predictions to give a policy which performs well for the contextual decision-making problem. This leads us to ask the following questions:

> *When can a black-box prediction model be turned into a contextual decision policy whose performance improves smoothly with prediction quality? Can such a policy be constructed efficiently, from both a statistical and computational point of view?*

1.1 Our Contributions

To address these questions we develop a framework for contextual optimization with black-box predictions and apply it to three natural problems: contextual posted pricing, ski rental, and house flipping. For all of these problems, we can formally show that FtP fails to perform well (Section 3.4), further motivating our approach. Our framework emphasizes the following three desiderata that any procedure for adapting a prediction model to downstream decisions ought to satisfy:

1. **Good predictions imply good decisions**: the procedure's decision loss should vanish as the prediction error vanishes.
2. **Statistically efficient**: the procedure should not require too much data to tune.
3. **Computationally efficient**: the procedure should add only minimal computational overhead.

To handle the first point, we propose a simple robustification of FtP. Instead of optimizing against the point prediction $h(x)$, we propose a class of robust policies $\{\pi_\varepsilon\}_{\epsilon\geq 0}$ which optimize against the worst case over an ε-neighborhood of $h(x)$. The parameter ε interpolates between fully trusting the prediction ($\varepsilon = 0$, which recovers FtP) and hedging against varying amounts of prediction error.

We show that the performance of this robust policy is governed by a single problem-specific quantity, the *robust optimization gap* $\Gamma(\varepsilon)$ (defined formally in Section 4), which measures the cost of hedging over an ε-ball. For any policy π, let $\mathrm{Loss}(\pi)$ denote the expected difference between the cost incurred by π and the optimal cost given the true state, where the expectation is taken over a joint distribution on context/state pairs (x, θ). If the prediction model has root-mean-squared error (RMSE) η, then we can show that for every $\varepsilon > 0$,

$$\mathrm{Loss}(\pi_\varepsilon) \leq \Gamma(\varepsilon) + \frac{M\eta^2}{\varepsilon^2},$$

where M bounds the per-instance loss. If $\Gamma(\varepsilon) = O(\varepsilon^\alpha)$, then choosing ε to balance the two terms shows that there exists a choice of ε so that $\mathrm{Loss}(\pi_\varepsilon) = O(\eta^{2\alpha/(\alpha+2)})$, which approaches 0 as $\eta \to 0$, satisfying our first desiderata. For all three application problems we can show that $\Gamma(\varepsilon) = O(\varepsilon)$, yielding a loss of $O(\eta^{2/3})$ for each.

The choice of ε above may depend on unknown quantities such as η or the distribution over context/state pairs. Thus we consider a data-driven approach to set ε where we are given samples $\{(x_i, \theta_i)\}_{i=1}^n$ of n context/state pairs. To satisfy the other two desiderata we require the method for tuning ε using the samples admit an efficient implementation and to quickly converge as the number of samples n increases. Via a simple empirical risk minimization (ERM) approach, for all three application problems we can give algorithms which output $\widehat{\varepsilon}$ so that with probability $1-\delta$, $\widehat{\pi} = \pi_{\widehat{\varepsilon}}$ satisfies

$$\mathrm{Loss}(\widehat{\pi}) \leq \inf_{\varepsilon\geq 0} \mathrm{Loss}(\pi_\varepsilon) + O\left(M\sqrt{\frac{\log n + \log\frac{1}{\delta}}{n}}\right),$$

for any $\delta \in (0, 1)$. Further, the algorithms are quite efficient, running in time $O(n \log n)$.

To contextualize our results further, it helps to consider the contextual posted pricing application. Medina and Vassilvitskii [18] consider this problem and use an elegant, clustering-based approach to

give a data-driven policy π satisfying $\mathrm{Loss}(\pi) \le O(\eta^{2/3}) + \tilde{O}((\log(1/\delta)/n)^{1/6})$ with probability $1-\delta$ over the choice of the samples. Combining our results end-to-end for posted pricing gives an improved bound over this prior work using an argument that we find substantially simpler.

1.2 Related Work

Algorithms with predictions. A recent literature studies how machine-learned predictions can improve classical online algorithms, with the goal of trading off performance under accurate predictions against robustness to mis-specification [19, 22]. Purohit et al. [22] and Gollapudi and Panigrahi [13] treat ski-rental and rent-or-buy with predictions, Bamas et al. [4] develop a primal-dual framework for learning-augmented online algorithms, and Wei and Zhang [24] characterize the optimal robustness-consistency trade-off. Anand et al. [1] study how to learn the predicted quantities themselves from contextual data, and Xu and Lu [25] initiate the study of mechanism design with predictions. Our setting differs from this literature on two counts: we study contextual stochastic decision problems with i.i.d. context–state pairs rather than adversarial online inputs, and we measure predictor quality via RMSE and decision quality via expected loss rather than competitive ratio.

Predict-then-optimize and prescriptive analytics. A second strand of literature studies how to use side information to predict unknown problem parameters and then optimize with respect to those predictions. Bertsimas and Kallus [5] formulate this perspective broadly under the heading of prescriptive analytics, and Elmachtoub and Grigas [12] develop the *smart predict-then-optimize* framework, in which the prediction model is trained to align with downstream decision quality. Our work is complementary: rather than redesigning the training objective or learning a decision rule end-to-end, we take the predictor as given and provide a deployment-time wrapper that converts it into a robust decision policy. This plug-in perspective applies when the predictor is expensive to train or is already fixed by the application.

Robust contextual optimization. Closer to our work are recent papers that augment contextual optimization with uncertainty sets around predictions. Sun et al. [23] propose a predict-then-calibrate paradigm for robust contextual linear programming, Patel et al. [21] develop a conformal framework for contextual robust optimization using nonconvex prediction regions, and Chenreddy and Delage [8] study end-to-end conditional robust optimization. These works and ours share the goal of using uncertainty quantification to improve downstream robustness. Our emphasis is different: we isolate a metric-ball robust policy around a point prediction, identify a single structural quantity $\Gamma(\varepsilon)$ that governs performance, and derive explicit prediction-dependent loss rates for several problems outside the LP setting whose costs have discontinuous, action-dependent thresholds.

Pricing, auctions, and side information. The closest application-specific antecedent is Medina and Vassilvitskii [18], who study reserve-price optimization with bid predictions and prove an $O(\eta^{2/3})$ upper bound together with a matching lower bound. Our pricing result recovers their upper bound through a substantially simpler minimax argument and then extends the same methodology to other applications. More broadly, there is a literature on pricing and auction design with side information, including feature-based dynamic pricing [9], sample-complexity guarantees for auctions with side information [11], and Myersonian regression for differentiated pricing [17]. Our goal is orthogonal to this literature: rather than designing a new pricing learner tailored to one setting, we develop a generic robustification principle that applies across multiple contextual decision problems.

1.3 Paper Organization

In Sections 2 and 3 we introduce our framework and formally define the three application problems. The remainder of the paper develops the general theory and then applies it to the three problems. Sections 4 and 5 introduce the robust policy and analyze its performance in terms of $\Gamma(\varepsilon)$. Section 6 considers data-driven selection of ε. Appendix A works out the three application problems in detail, deriving the closed-form robust policies and their loss bounds. Finally, Appendix B gives further discussion of the robust optimization gap $\Gamma(\varepsilon)$.

2 Problem Setup

We consider stochastic contextual optimization with a fixed black-box prediction model. An instance consists of a context space $\mathcal{X}$, a state space Θ equipped with a metric ρ, an action space $\mathcal{A}$, an unknown distribution $\mathcal{D}$ over $\mathcal{X} \times \Theta$, and a cost function $\mathrm{Cost} : \mathcal{A} \times \Theta \to \mathbb{R}$. The decision-maker observes a context x, chooses an action a, and incurs cost $\mathrm{Cost}(a, \theta)$ for the realized state θ.

We assume a prediction model $h : \mathcal{X} \to \Theta$ is given. We measure its quality by its root-mean-squared error

$$\eta := \sqrt{\mathbb{E}_{(x,\theta)\sim\mathcal{D}}\left[\rho(h(x), \theta)^2\right]}.$$

A policy $\pi : \mathcal{X} \to \mathcal{A}$ maps contexts to actions. Let

$$\mathrm{Opt}(\theta) := \inf_{a\in\mathcal{A}} \mathrm{Cost}(a, \theta)$$

denote the clairvoyant optimum in state θ. The loss of a policy is its expected excess cost relative to this clairvoyant benchmark:

$$\mathrm{Loss}(\pi) := \mathbb{E}_{(x,\theta)\sim\mathcal{D}}\left[\mathrm{Cost}(\pi(x), \theta) - \mathrm{Opt}(\theta)\right].$$

Goal. Given a black-box predictor h, construct h-dependent policies π^h whose loss vanishes with the prediction error:

$$\mathrm{Loss}(\pi^h) \le f(\eta) \qquad \text{for some } f \text{ satisfying } f(\eta) \to 0 \text{ as } \eta \to 0.$$

We call such policies η-*sensitive*.

3 Three Applications

We instantiate the framework of Section 2 on three contextual decision problems that will serve as running examples throughout the paper. The first two are standard models from pricing and online algorithms; the third is a stylized example designed to illustrate that the framework also applies when the payoff-relevant state is multidimensional. In each case we specify the state space Θ, action space $\mathcal{A}$, cost function Cost, metric ρ, and clairvoyant optimum Opt.

3.1 Posted Pricing

Posted-pricing problems are fundamental in revenue management and online learning. The prediction-based setting is especially close to Medina and Vassilvitskii [18], who relate revenue guarantees to the quality of a standard squared-loss predictor. In this problem, a seller posts a price p to a buyer with private value $v \in [0, 1]$. The buyer purchases iff $v \ge p$, yielding revenue p. The model is

$$\Theta = [0, 1],\ \mathcal{A} = [0, 1] \qquad \rho(v, w) = |v - w|$$
$$\mathrm{Cost}(p, v) = -p\,\mathbf{1}\{v \ge p\} \qquad \mathrm{Opt}(v) = -v$$

The optimum is attained by posting $p = v$.

3.2 Ski Rental

Ski rental is the classical rent-or-buy problem in online algorithms: the decision-maker must decide when to pay a large one-time cost in order to avoid continued rental costs [16]. It has also become a standard testbed for algorithms with machine-learned predictions [22]. In this problem, b is the cost of buying skis while renting costs 1 per unit time. We assume a known upper bound $L \ge b$ such that the trip length ℓ lies in $[0, L]$, and the decision-maker chooses a time $t \in [0, L]$ at which to buy. The model is

$$\Theta = [0, L],\ \mathcal{A} = [0, L] \qquad \rho(\ell, \ell') = |\ell - \ell'|$$
$$\mathrm{Cost}(t, \ell) = (t + b)\,\mathbf{1}\{t < \ell\} + \ell\,\mathbf{1}\{t \ge \ell\} \qquad \mathrm{Opt}(\ell) = \min\{\ell, b\}$$

The clairvoyant policy rents for the entire trip if $\ell < b$ and buys immediately if $\ell \ge b$.

3.3 House Flipping

This example illustrates a problem with a multidimensional state space. In house flipping, an investor considers buying a house with unknown current market value v and unknown future resale value v', both in $[0, V]$ for a known $V > 0$. The investor makes an offer p; if $p \geq v$, the offer is accepted and the house is later sold for v', yielding net profit $v' - p$. The model is

$$\Theta = [0,V]^2,\ \mathcal{A} = [0,V] \qquad \rho\big((v,v'),(w,w')\big) = \max\{|v-w|, |v'-w'|\}$$
$$\mathrm{Cost}\big(p,(v,v')\big) = -(v'-p)\,\mathbf{1}\{p \geq v\} \qquad \mathrm{Opt}\big((v,v')\big) = -\max\{v'-v, 0\}$$

If $v' > v$, the optimum is attained by offering $p = v$; otherwise, the investor optimally avoids purchasing.

Here the state is two-dimensional, so we write the predictor as $h = (h_v, h_{v'}) : \mathcal{X} \to [0,V]^2$. The joint RMSE under the ℓ^∞ metric is

$$\eta := \sqrt{\mathbb{E}_{(x,(v,v'))\sim\mathcal{D}}\Big[\max\big(|h_v(x) - v|,\ |h_{v'}(x) - v'|\big)^2\Big]}.$$

If η_v and $\eta_{v'}$ denote the individual RMSEs for the two coordinates, then

$$\eta \leq \sqrt{\eta_v^2 + \eta_{v'}^2},$$

so a bound on the joint RMSE follows from bounds on the individual prediction errors.

3.4 Follow-the-Prediction Is Not η-Sensitive

Having defined the three problems, we observe that the follow-the-prediction (FtP) baseline (which best-responds to $h(x)$ as if it were the true state), is not η-sensitive on any of them. This failure persists even when each coordinate of $h(x)$ is the conditional mean, $\mathbb{E}[\theta_i \mid x]$, which is the best possible coordinatewise predictor in the mean-squared sense.

Proposition 3.1 (FtP is not η-sensitive)**.** *For each of the three applications above, there exists a context space $\mathcal{X}$, a distribution $\mathcal{D}$, and a minimum-variance predictor h with RMSE $\eta > 0$ such that*

$$\mathrm{Loss}(\pi_{\mathrm{FtP}}) \geq c$$

for a constant $c > 0$ independent of η. Explicitly, one may take $c = 1/4$ for pricing (for any $\eta \leq 1/2$), $c = b/4$ for ski-rental (for any $\eta \leq b/2$), and $c = C/2$ for house flipping (for any fixed $C \in (0, V/2]$, any $\eta \leq C$).

Proof for contextual pricing. Take $\mathcal{X} = \{x_0\}$ a singleton, set $h(x_0) = 1/2$, and draw $v \mid x_0 = 1/2 + \eta\epsilon$ where ϵ is an independent Rademacher random variable. Then $\mathbb{E}[v \mid x_0] = 1/2 = h(x_0)$, so h is a minimum-variance predictor, and $\mathbb{E}[(v - h(x_0))^2] = \eta^2$. The FtP policy posts $p = h(x_0) = 1/2$. A sale occurs iff $v \geq p$, iff $\epsilon = +1$, which happens with probability $1/2$. Hence

$$\mathrm{Loss}(\pi_{\mathrm{FtP}}) = \mathbb{E}[v] - p \cdot \Pr[v \geq p] = \tfrac{1}{2} - \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4},$$

independent of η. The proofs for the other applications are analogous and deferred to Appendix C. □

More broadly, we show that in general there is no single rule using only the predictor that is η-sensitive. Thus additional information, such as knowledge of η or calibration data, is necessary. We prove the following proposition in Appendix D using similar constructions as in Proposition 3.1.

Proposition 3.2 (Impossibility for fixed predictor-only policies)**.** *For each of contextual pricing and house flipping, there is a fixed predictor h and a family of distributions $(\mathcal{D}_\eta)_{0<\eta<\eta_0}$ for some $\eta_0 > 0$, with RMSE η, such that every possibly randomized policy π^h, fixed independently of η and $\mathcal{D}_\eta$, satisfies $\liminf_{\eta\downarrow 0} \mathrm{Loss}_{\mathcal{D}_\eta}(\pi^h) > 0$.*

4 The Robust Policy

For the robust policy, rather than best-responding to $h(x)$ directly, we protect against the worst case within a neighborhood of the prediction. We consider a family of policies indexed by a radius ε, which can be chosen using prediction-error information or calibration data.

Definition 4.1 (Minimax Value)**.** *For $\varepsilon \geq 0$ and $\theta_0 \in \Theta$, the minimax value is*

$$\mathrm{Val}(\varepsilon, \theta_0) := \inf_{a \in \mathcal{A}} \sup_{\theta \in B(\theta_0, \varepsilon)} \mathrm{Cost}(a, \theta),$$

where $B(\theta_0, \varepsilon) := \{\theta \in \Theta : \rho(\theta_0, \theta) \leq \varepsilon\}$ is the closed ε-ball around θ_0.

Definition 4.2 (Robust Policy)**.** *For $\varepsilon \geq 0$, the robust policy $\pi_\varepsilon : \mathcal{X} \to \mathcal{A}$ is defined by*

$$\pi_\varepsilon(x) \in \arg\min_{a \in \mathcal{A}} \sup_{\theta \in B(h(x), \varepsilon)} \mathrm{Cost}(a, \theta),$$

so that $\sup_{\theta \in B(h(x), \varepsilon)} \mathrm{Cost}(\pi_\varepsilon(x), \theta) = \mathrm{Val}(\varepsilon, h(x))$. We assume throughout that this argmin is attained.[1]

Remark 4.3 (Robust Policy Generalizes Follow-the-Prediction)**.** *At $\varepsilon = 0$, the ball $B(h(x), 0) = \{h(x)\}$ is a singleton, so the robust policy reduces to*

$$\pi_0(x) = \arg\min_{a \in \mathcal{A}} \mathrm{Cost}(a, h(x)),$$

which is the best-response to the prediction, i.e., the follow-the-prediction (FtP) policy. Thus the family $\{\pi_\varepsilon : \varepsilon \geq 0\}$ contains FtP as a special case, and we may write $\pi_{\mathrm{FtP}} = \pi_0$.

Compared to follow-the-prediction (which amounts to solving a *nominal* optimization problem), one may expect that implementing the robust policy incurs significant additional complexity. For all three applications, we derive closed-form robust policies in Appendix A. Appendix E discusses efficient implementations for contextual linear optimization.

Our main goal is to understand the loss of the robust policy, which we connect to two quantities: the *robust optimization gap*, which controls the loss when the prediction is accurate, and the *maximum loss*, which bounds the loss in the worst case.

Definition 4.4 (Robust Optimization Gap)**.** *For $\varepsilon \geq 0$, the robust optimization gap is*

$$\Gamma(\varepsilon) := \sup_{\theta_0 \in \Theta} \sup_{\theta \in B(\theta_0, \varepsilon)} \left[\mathrm{Val}(\varepsilon, \theta_0) - \mathrm{Opt}(\theta)\right].$$

In words, $\Gamma(\varepsilon)$ is the worst-case gap between the cost of the minimax-optimal action over an ε-ball and the clairvoyant optimum at any state in that ball. It measures the price of hedging: how much we lose by protecting against ε-perturbations rather than acting with full knowledge of the true state.

Definition 4.5 (Maximum Loss)**.** *The maximum loss is*

$$M := \sup_{\varepsilon \geq 0,\, x \in \mathcal{X},\, \theta \in \Theta} \left[\mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta)\right].$$

We assume $M < \infty$. This bounds the pointwise loss of the entire robust family. For example, uniformly bounded costs $|\mathrm{Cost}(a, \theta)| \leq K$ imply $M \leq 2K$.

Remark 4.6. *The robust optimization gap satisfies several natural properties:*

- *(i) $\Gamma(\varepsilon) \geq 0$ for all $\varepsilon \geq 0$, since $\mathrm{Val}(\varepsilon, \theta_0) \geq \mathrm{Opt}(\theta)$ whenever $\theta \in B(\theta_0, \varepsilon)$.*
- *(ii) Γ is non-decreasing: larger balls lead to a (weakly) larger gap.*
- *(iii) Γ depends only on Cost, $\mathcal{A}$, Θ, and ρ, not on the distribution $\mathcal{D}$ or the prediction model h.*

5 General Loss Bound

We now present the main results. The first lemma establishes a point-wise loss bound for the robust policy, which is the key structural result.

Lemma 5.1 (Point-wise Loss Bound)**.** *For all $\varepsilon \geq 0$, $x \in \mathcal{X}$, and $\theta \in \Theta$,*

$$\mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta) \leq \Gamma(\varepsilon) \cdot \mathbf{1}\{\rho(h(x), \theta) \leq \varepsilon\} + M \cdot \mathbf{1}\{\rho(h(x), \theta) > \varepsilon\}.$$

[1] This holds under standard compactness and continuity conditions, and is verified directly for each application we consider.

Proof. We consider two cases.

Case 1: $\rho(h(x), \theta) \le \varepsilon$**.** Since $\theta \in B(h(x), \varepsilon)$, we have

$$\mathrm{Cost}(\pi_\varepsilon(x), \theta) \le \sup_{\theta' \in B(h(x),\varepsilon)} \mathrm{Cost}(\pi_\varepsilon(x), \theta') = \mathrm{Val}(\varepsilon, h(x)).$$

By Definition 4.4 with $\theta_0 = h(x)$ and $\theta \in B(h(x), \varepsilon)$,

$$\mathrm{Val}(\varepsilon, h(x)) - \mathrm{Opt}(\theta) \le \Gamma(\varepsilon).$$

Case 2: $\rho(h(x), \theta) > \varepsilon$**.** By Definition 4.5, $\mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta) \le M$. □

Taking expectations yields the general loss bound.

Theorem 5.2 (General Loss Bound)**.** *For any* $\varepsilon > 0$*,*

$$\mathrm{Loss}(\pi_\varepsilon) \le \Gamma(\varepsilon) + \frac{M\eta^2}{\varepsilon^2}.$$

Proof. Taking expectations over $(x, \theta) \sim \mathcal{D}$ in Lemma 5.1,

$$\begin{aligned}\mathrm{Loss}(\pi_\varepsilon) &\le \Gamma(\varepsilon) \cdot \Pr[\rho(h(x), \theta) \le \varepsilon] + M \cdot \Pr[\rho(h(x), \theta) > \varepsilon] \\ &\le \Gamma(\varepsilon) + M \cdot \Pr[\rho(h(x), \theta)^2 > \varepsilon^2] \le \Gamma(\varepsilon) + M \cdot \frac{\mathbb{E}[\rho(h(x), \theta)^2]}{\varepsilon^2} = \Gamma(\varepsilon) + \frac{M\eta^2}{\varepsilon^2},\end{aligned}$$

where the second line uses that probabilities are at most 1, and the third applies Markov's inequality. □

We define the optimal radius as the one that minimizes the loss.

Definition 5.3 (Optimal radius)**.** *The* optimal radius *is*

$$\varepsilon^* \in \arg\min_{\varepsilon \ge 0} \mathrm{Loss}(\pi_\varepsilon),$$

and $\mathrm{Loss}(\pi_{\varepsilon^*}) = \min_{\varepsilon \ge 0} \mathrm{Loss}(\pi_\varepsilon)$ *is the loss of the best policy in the robust family.*

When $\Gamma(\varepsilon)$ has a power-law form, the loss at the optimal radius admits a clean bound.

Corollary 5.4 (Loss at the optimal radius)**.** *Suppose* $\Gamma(\varepsilon) \le C\varepsilon^\alpha$ *for constants* $C, \alpha > 0$*. Then*

$$\mathrm{Loss}(\pi_{\varepsilon^*}) \le 2C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} \eta^{2\alpha/(\alpha+2)}.$$

Proof. Let $\varepsilon_0 := (M\eta^2/C)^{1/(\alpha+2)}$. By Theorem 5.2,

$$\mathrm{Loss}(\pi_{\varepsilon_0}) \le C\varepsilon_0^\alpha + \frac{M\eta^2}{\varepsilon_0^2} = C\left(\frac{M\eta^2}{C}\right)^{\alpha/(\alpha+2)} + M\eta^2\left(\frac{C}{M\eta^2}\right)^{2/(\alpha+2)}.$$

Both terms simplify to $C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} \eta^{2\alpha/(\alpha+2)}$, giving $\mathrm{Loss}(\pi_{\varepsilon_0}) \le 2C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} \eta^{2\alpha/(\alpha+2)}$. Since ε^* minimizes the loss over $\varepsilon \ge 0$, we have $\mathrm{Loss}(\pi_{\varepsilon^*}) \le \mathrm{Loss}(\pi_{\varepsilon_0})$, completing the proof. □

Beyond the three applications. The framework also applies to inventory with a fixed stockout fee. For demand θ and order a in $[0, D]$, let

$$\mathrm{Cost}(a, \theta) = c_o(a - \theta)^+ + c_u(\theta - a)^+ + \kappa \mathbf{1}\{\theta > a\}, \qquad c_o, c_u, \kappa > 0.$$

This cost is discontinuous and nonmonotone in demand. FtP orders $h(x)$ and incurs the fixed fee with probability $1/2$ on demands $h(x) \pm \eta$, even as $\eta \to 0$. Nevertheless, ordering the upper endpoint of the uncertainty interval avoids the fee and incurs at most $2c_o\varepsilon$, so $\Gamma(\varepsilon) \le 2c_o\varepsilon$. Since losses are bounded, Corollary 5.4 applies. Appendix B.5 gives the exact robust policy and learning guarantees; Appendix B.4 gives a coupled multi-item example with multidimensional states and actions. Appendix B gives general conditions for a vanishing robust gap.

Since ε^* is defined in terms of the unknown distribution, both ε^* and η are unobservable in practice. Section 6 develops procedures for selecting the radius from data, starting with the simplest MSE plug-in and progressing to selectors with stronger guarantees.

6 Data-Driven Selection of ε

We assume that we are given n independent and identically distributed (IID) samples $\{(x_i, \theta_i)\}_{i=1}^n$ from $\mathcal{D}$ in addition to the fixed predictor h. Throughout this section, h is fixed independently of the IID calibration sample, for example by training on a separate split. Proposition 3.2 shows that achieving η-sensitivity in general requires information beyond the predictor. Here, we obtain this information from calibration data.

A pricing example. In the pricing example of Proposition 3.1, the robust price is $\max\{1/2 - \varepsilon, 0\}$. For $0 < \eta < 1/2$, an optimal radius is η when $\eta < 1/4$ and 0 when $\eta > 1/4$; either is optimal at equality. The resulting loss is $\min\{\eta, 1/4\}$. For small η, choosing $\varepsilon = \eta$ reduces the loss from $1/4$ under FtP to η. Appendix D.2 gives the full calculation and shows how ERM learns the choice from samples.

The simplest data-driven way to select ε is to plug in an empirical MSE estimate of η in place of the unknown true value, using the same closed-form balancing as in the proof of Corollary 5.4.

Theorem 6.1 (Sample-Based Loss Bound)**.** *Suppose $\Gamma(\varepsilon) \le C\varepsilon^\alpha$ for constants $C, \alpha > 0$ and that $\rho(h(x), \theta)^2 \le B$ almost surely for some constant $B > 0$. Given n IID samples $\{(x_i, \theta_i)\}_{i=1}^n$ from $\mathcal{D}$ and $\delta \in (0, 1)$, define the empirical MSE*

$$\hat{\eta}^2 := \frac{1}{n} \sum_{i=1}^n \rho(h(x_i), \theta_i)^2,$$

and set $\gamma := B\sqrt{\log(2/\delta)/(2n)}$ and $\hat{\varepsilon}_{\mathrm{mse}} := (M(\hat{\eta}^2 + \gamma)/C)^{1/(\alpha+2)}$. Then, with probability at least $1 - \delta$,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{mse}}}) \le 2C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} \eta^{2\alpha/(\alpha+2)} + C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} (2\gamma)^{\alpha/(\alpha+2)}.$$

In particular, the second term is $O\left(\left(\frac{\log(1/\delta)}{n}\right)^{\alpha/(2(\alpha+2))}\right)$ with constants depending on B, C, and M.

Proof. Since $\rho(h(x_i), \theta_i)^2 \in [0, B]$ are IID with mean η^2, Hoeffding's inequality gives $\Pr[|\hat{\eta}^2 - \eta^2| > \gamma] \le \delta$. Condition on the event $|\hat{\eta}^2 - \eta^2| \le \gamma$, which holds with probability at least $1 - \delta$.

By Theorem 5.2,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{mse}}}) \le C\hat{\varepsilon}_{\mathrm{mse}}^\alpha + \frac{M\eta^2}{\hat{\varepsilon}_{\mathrm{mse}}^2}.$$

We bound each term.

First term. On the good event, $\hat{\eta}^2 + \gamma \le \eta^2 + 2\gamma$, so

$$C\hat{\varepsilon}_{\mathrm{mse}}^\alpha = C\left(\frac{M}{C}\right)^{\alpha/(\alpha+2)} (\hat{\eta}^2 + \gamma)^{\alpha/(\alpha+2)} \le C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} (\eta^2 + 2\gamma)^{\alpha/(\alpha+2)}.$$

Since $\alpha/(\alpha+2) \in (0, 1)$, the function $t \mapsto t^{\alpha/(\alpha+2)}$ is subadditive on $[0, \infty)$, so

$$(\eta^2 + 2\gamma)^{\alpha/(\alpha+2)} \le \eta^{2\alpha/(\alpha+2)} + (2\gamma)^{\alpha/(\alpha+2)}.$$

Second term. Let $\varepsilon_0 := (M\eta^2/C)^{1/(\alpha+2)}$ be the bound-minimizing radius from the proof of Corollary 5.4. On the good event, $\hat{\eta}^2 + \gamma \ge \eta^2$, so $\hat{\varepsilon}_{\mathrm{mse}} \ge \varepsilon_0$ and thus

$$\frac{M\eta^2}{\hat{\varepsilon}_{\mathrm{mse}}^2} \le \frac{M\eta^2}{\varepsilon_0^2} = C^{2/(\alpha+2)} M^{\alpha/(\alpha+2)} \eta^{2\alpha/(\alpha+2)}.$$

Combining both terms yields the result. □

Theorem 6.1 achieves the rate of Corollary 5.4 up to an $O(n^{-\alpha/(2(\alpha+2))})$ statistical term. Its proof rests on two upper-bound steps: Markov's inequality on the squared prediction error and the closed-form balancing choice of $\hat{\varepsilon}_{\mathrm{mse}}$. We now develop two improved selectors:

- **Empirical CDF** (Section 6.1) replaces Markov by direct estimation of the tail $\Pr[\rho(h(x), \theta) > \varepsilon]$ via DKW. The resulting selector competes with $\mathrm{Loss}_{\mathrm{CDF}} := \min_{\varepsilon}[\Gamma(\varepsilon) + MF(\varepsilon)]$, an upper bound on $\mathrm{Loss}(\pi_{\varepsilon^*})$.
- **Empirical risk minimization** (Section 6.2) bypasses the closed-form balancing by minimizing the empirical loss over $\{\pi_\varepsilon : \varepsilon \geq 0\}$ directly. The selector competes with $\mathrm{Loss}(\pi_{\varepsilon^*})$ directly, and dominates FtP since $\pi_0 = \pi_{\mathrm{FtP}}$.

6.1 Direct Tail Estimation via the Empirical CDF

The Markov step in the proof of Theorem 6.1 bounds the tail $F(\varepsilon) := \Pr[\rho(h(x), \theta) > \varepsilon]$ indirectly, via the squared prediction error. This is lossy: it reduces the entire error distribution to a single number η, discarding distributional information. A more direct approach is to work with the tail function F itself.

As noted above, the general loss bound takes the form

$$\mathrm{Loss}(\pi_\varepsilon) \leq \Gamma(\varepsilon) + M \cdot F(\varepsilon).$$

If F were known, the optimal choice would be $\varepsilon_{\mathrm{CDF}} := \arg\min_{\varepsilon>0}[\Gamma(\varepsilon) + M \cdot F(\varepsilon)]$, achieving the *CDF bound*

$$\mathrm{Loss}_{\mathrm{CDF}} := \min_{\varepsilon>0} \left[\Gamma(\varepsilon) + M \cdot F(\varepsilon)\right].$$

This CDF bound is at least as tight as the Markov-based bound of Theorem 6.1, since Markov's inequality $F(\varepsilon) \leq \eta^2/\varepsilon^2$ only provides an upper bound on F. For light-tailed error distributions (e.g., sub-Gaussian), where F decays exponentially rather than polynomially, the CDF bound can be substantially smaller.

To achieve this in practice, we estimate F from samples using the empirical survival function

$$\hat{F}(\varepsilon) := \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{\rho(h(x_i), \theta_i) > \varepsilon\}.$$

The Dvoretzky–Kiefer–Wolfowitz (DKW) inequality provides a *uniform* confidence band: with probability at least $1 - \delta$,

$$\sup_{\varepsilon \geq 0} \left| F(\varepsilon) - \hat{F}(\varepsilon) \right| \leq \gamma, \quad \text{where } \gamma := \sqrt{\frac{\log(2/\delta)}{2n}}.$$

We then set $\hat{\varepsilon}_{\mathrm{cdf}} := \arg\min_{\varepsilon>0}[\Gamma(\varepsilon) + M\hat{F}(\varepsilon)]$.

Proposition 6.2. *With probability at least $1 - \delta$ over the sample,*

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{cdf}}}) \leq \mathrm{Loss}_{\mathrm{CDF}} + 2M\gamma = \min_{\varepsilon>0} \left[\Gamma(\varepsilon) + M \cdot F(\varepsilon)\right] + 2M\sqrt{\frac{\log(2/\delta)}{2n}}.$$

Proof. Condition on the DKW good event $|F(\varepsilon) - \hat{F}(\varepsilon)| \leq \gamma$ for all ε. By the first inequality in the proof of Theorem 5.2 (before applying Markov's inequality):

$$\begin{aligned}
\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{cdf}}}) &\leq \Gamma(\hat{\varepsilon}_{\mathrm{cdf}}) + M \cdot F(\hat{\varepsilon}_{\mathrm{cdf}}) \\
&\leq \Gamma(\hat{\varepsilon}_{\mathrm{cdf}}) + M(\hat{F}(\hat{\varepsilon}_{\mathrm{cdf}}) + \gamma) \leq \Gamma(\varepsilon_{\mathrm{CDF}}) + M(\hat{F}(\varepsilon_{\mathrm{CDF}}) + \gamma) \\
&\leq \Gamma(\varepsilon_{\mathrm{CDF}}) + M(F(\varepsilon_{\mathrm{CDF}}) + 2\gamma) = \mathrm{Loss}_{\mathrm{CDF}} + 2M\gamma,
\end{aligned}$$

where the third line uses the optimality of $\hat{\varepsilon}_{\mathrm{cdf}}$ for the empirical criterion, and the second and fourth lines use the DKW bound. □

A further advantage is that the approach is fully adaptive: the empirical optimization over ε automatically selects the best tradeoff between Γ and F.

6.2 Empirical Risk Minimization and Dominance over FtP

Whereas the DKW selector competes with $\mathrm{Loss}_{\mathrm{CDF}}$, we now show that minimizing the empirical loss directly yields a selector that competes with $\mathrm{Loss}(\pi_{\varepsilon^*})$, and as a consequence dominates FtP since $\pi_0 = \pi_{\mathrm{FtP}}$ lies in the family $\{\pi_\varepsilon : \varepsilon \geq 0\}$ (Remark 4.3).

Given n IID samples $\{(x_i, \theta_i)\}_{i=1}^n$ from $\mathcal{D}$, define the empirical loss

$$\widehat{L}(\varepsilon) := \frac{1}{n} \sum_{i=1}^{n} \left[\mathrm{Cost}(\pi_\varepsilon(x_i), \theta_i) - \mathrm{Opt}(\theta_i)\right],$$

and set $\hat{\varepsilon}_{\mathrm{erm}} := \arg\min_{\varepsilon \geq 0} \widehat{L}(\varepsilon)$.

Recall that the *pseudo-dimension* of a class $\mathcal{F}$ of real-valued functions on a domain $\mathcal{Z}$ is the largest integer d such that there exist $z_1, \ldots, z_d \in \mathcal{Z}$ and thresholds $t_1, \ldots, t_d \in \mathbb{R}$ with the property that for every subset $S \subseteq \{1, \ldots, d\}$, there exists $f \in \mathcal{F}$ satisfying $f(z_i) \geq t_i$ if and only if $i \in S$. It generalizes the VC-dimension to real-valued function classes and controls the rate of uniform convergence [3, 14].

Theorem 6.3 (Excess Loss Bound for ERM)**.** *Suppose* $\mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta) \in [0, M]$ *for all* x, θ, ε*. Let* d *denote the pseudo-dimension of the function class* $\mathcal{F} = \{\ell_\varepsilon : \varepsilon \geq 0\}$*, where* $\ell_\varepsilon(x, \theta) := \mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta)$*. If* $d < \infty$*, then with probability at least* $1 - \delta$ *over the sample,*

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq \mathrm{Loss}(\pi_{\varepsilon^*}) + O\left(M\sqrt{\frac{d\log(n) + \log(1/\delta)}{n}}\right).$$

In particular, since $\pi_0 = \pi_{\mathrm{FtP}}$*,*

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq \mathrm{Loss}(\pi_{\mathrm{FtP}}) + O\left(M\sqrt{\frac{d\log(n) + \log(1/\delta)}{n}}\right).$$

Proof. By standard uniform convergence for bounded function classes with finite pseudo-dimension (see, e.g., [3, 14]), with probability at least $1 - \delta$,

$$\sup_{\varepsilon \geq 0} \left|\widehat{L}(\varepsilon) - \mathrm{Loss}(\pi_\varepsilon)\right| \leq c,$$

where $c = O\left(M\sqrt{\frac{d\log(n)+\log(1/\delta)}{n}}\right)$.

On this event, for any $\varepsilon_0 \geq 0$,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq \widehat{L}(\hat{\varepsilon}_{\mathrm{erm}}) + c \leq \widehat{L}(\varepsilon_0) + c \leq \mathrm{Loss}(\pi_{\varepsilon_0}) + 2c,$$

where the second line uses the optimality of $\hat{\varepsilon}_{\mathrm{erm}}$ for the empirical criterion. Taking $\varepsilon_0 = \varepsilon^*$ gives the first bound; taking $\varepsilon_0 = 0$ gives the second. □

In Appendix A.4 we show that the pseudo-dimension is constant for each of the three applications.

Computing the radius. The closed-form policies for the three applications also make radius selection efficient (Propositions A.1, A.5 and A.8). Their per-sample losses are piecewise affine in ε with only a constant number of breakpoints. An $O(n \log n)$ calibration algorithm sorts these breakpoints and sweeps the empirical loss, accounting for exact endpoint values and one-sided limits (Proposition A.13). The time bound excludes evaluating the given predictor.

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A Applications

We now apply the framework to the three problems of Section 3. In contrast to the FtP failures of Proposition 3.1, the robust policy achieves $O(\eta^{2/3})$ loss rates in all three settings. For each problem, we (1) derive the robust policy π_ε by solving the minimax problem explicitly, (2) compute the robust optimization gap $\Gamma(\varepsilon)$ and maximum loss M, and (3) state the resulting loss bound.

A.1 Contextual Pricing

Recall the contextual pricing problem from Section 3.1.

Proposition A.1 (Robust Policy for Pricing)**.** *For the contextual pricing problem, the robust policy is*

$$\pi_\varepsilon(x) = \max\{h(x) - \varepsilon,\, 0\}.$$

Proof. We solve $\min_{p\in[0,1]} \max_{v\in B(h(x),\varepsilon)\cap[0,1]} \mathrm{Cost}(p, v)$. Since $\mathrm{Cost}(p, v) = -p \cdot \mathbf{1}\{v \geq p\}$ is non-increasing in v for fixed p, the inner maximum is attained at $v = \max\{h(x) - \varepsilon, 0\}$. Thus the robust policy reduces to the best response against $v = \max\{h(x) - \varepsilon, 0\}$, which sets $p^* = \max\{h(x) - \varepsilon, 0\}$. □

Proposition A.2 (Robust Optimization Gap for Pricing)**.** *For the contextual pricing problem,* $\Gamma(\varepsilon) \leq 2\varepsilon$ *and* $M = 1$.

Proof. From the proof of Proposition A.1, $\mathrm{Val}(\varepsilon, v_0) = -\max\{v_0 - \varepsilon, 0\}$. Since $\mathrm{Opt}(v) = -v$, the gap is

$$\mathrm{Val}(\varepsilon, v_0) - \mathrm{Opt}(v) = v - \max\{v_0 - \varepsilon, 0\} \leq v - (v_0 - \varepsilon) = (v - v_0) + \varepsilon \leq 2\varepsilon,$$

using $\max\{v_0 - \varepsilon, 0\} \geq v_0 - \varepsilon$ and $v \leq v_0 + \varepsilon$. Equality holds at $v = v_0 + \varepsilon$ when $v_0 \geq \varepsilon$, so the bound is tight for $\varepsilon \leq 1/2$.

For M: $\mathrm{Cost}(p, v) - \mathrm{Opt}(v) = v - p \cdot \mathbf{1}\{v \geq p\} \leq v \leq 1$, so $M = 1$. □

Corollary A.3 (Pricing Loss Bound)**.** *Applying Corollary 5.4 with* $C = 2$, $\alpha = 1$, *and* $M = 1$ *gives*

$$\mathrm{Loss}(\pi_{\varepsilon^*}) \leq 2^{5/3}\, \eta^{2/3}.$$

Combined with the finite pseudo-dimension established in Proposition A.11 below, the ERM selector of Theorem 6.3 gives, with probability at least $1 - \delta$,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq 2^{5/3}\, \eta^{2/3} + O\left(\sqrt{\frac{\log n + \log(1/\delta)}{n}}\right).$$

The $O(\eta^{2/3})$ rate in Corollary A.3 is optimal.

Theorem A.4 (Pricing Lower Bound)**.** *For any* $\eta \in (0, 1)$ *and any algorithm* A *that maps a prediction model* $h : \mathcal{X} \to [0, 1]$ *to a policy* $\pi_A^h : \mathcal{X} \to [0, 1]$, *there exists a distribution* $\mathcal{D}$ *over* $\mathcal{X} \times [0, 1]$ *and a prediction model* h *with* $\mathrm{RMSE}(h) \leq \eta$ *such that*

$$\mathrm{Loss}(\pi_A^h) \geq \eta^{2/3}(1 - \eta^{2/3}).$$

Let $\mathrm{Loss}^*(\eta) := \inf_A \sup_{(\mathcal{D},h):\, \mathrm{RMSE}(h)\leq\eta} \mathrm{Loss}(\pi_A^h)$ *denote the minimax loss over predictor-aware algorithms* A *and over instances* $(\mathcal{D}, h)$ *with RMSE at most* η. *Together with Corollary A.3, this gives*

$$\eta^{2/3}(1 - \eta^{2/3}) \leq \mathrm{Loss}^*(\eta) \leq 2^{5/3}\, \eta^{2/3},$$

so the minimax loss for contextual pricing scales as $\Theta(\eta^{2/3})$.

Proof. Consider the trivial instance $\mathcal{X} = \{x_0\}$ with prediction $h(x_0) = 1$ and the two-point distribution

$$v = \begin{cases} 1 & \text{with probability } 1 - \eta^{2/3}, \\ 1 - \eta^{2/3} & \text{with probability } \eta^{2/3}. \end{cases}$$

The RMSE is $\sqrt{\eta^{2/3} \cdot (\eta^{2/3})^2} = \eta$, and the expected optimal revenue is $\mathbb{E}[v] = 1 - \eta^{4/3}$. Since $\mathcal{X}$ is a singleton and h is fixed, the algorithm's output reduces to a single price $p := \pi_A^h(x_0) \in [0, 1]$.

Case 1: $p \leq 1 - \eta^{2/3}$**.** Both types purchase, so the loss is $\mathbb{E}[v] - p \geq (1 - \eta^{4/3}) - (1 - \eta^{2/3}) = \eta^{2/3}(1 - \eta^{2/3})$.

Case 2: $p > 1 - \eta^{2/3}$**.** The low type ($v = 1 - \eta^{2/3}$) does not purchase, so the revenue is $p \cdot \Pr[v \geq p] \leq p(1-\eta^{2/3}) \leq 1-\eta^{2/3}$ and the loss is $\mathbb{E}[v]-p\cdot\Pr[v \geq p] \geq (1-\eta^{4/3})-(1-\eta^{2/3}) = \eta^{2/3}(1 - \eta^{2/3})$. □

A.2 Contextual Ski-Rental

Recall the contextual ski-rental problem from Section 3.2.

Proposition A.5 (Robust Policy for Ski-Rental)**.** *For the contextual ski-rental problem, the robust policy is*

$$\pi_\varepsilon(x) = \begin{cases} (h(x)+\varepsilon)\cdot \mathbf{1}\{h(x)+\varepsilon < b\}, & \varepsilon > 0, \\ L \cdot \mathbf{1}\{h(x) < b\}, & \varepsilon = 0. \end{cases}$$

Proof. We solve $\min_{t\in[0,L]} \max_{\ell\in B(h(x),\varepsilon)\cap[0,L]} \mathrm{Cost}(t,\ell)$. Since $\mathrm{Cost}(t,\ell)$ is non-decreasing in ℓ for fixed t, the inner maximum is attained at $u_\varepsilon := \min\{h(x)+\varepsilon, L\}$. For a known trip length u_ε, it is optimal either to buy immediately ($t=0$, cost $= b$) or to rent for the entire trip ($t \geq u_\varepsilon$, cost $= u_\varepsilon$), so the minimax value is $\mathrm{Val}(\varepsilon, h(x)) = \min(u_\varepsilon, b)$. For $\varepsilon > 0$, if $u_\varepsilon < b$ then $u_\varepsilon = h(x)+\varepsilon$ (since $L \geq b$), and we tie-break by choosing $t = h(x)+\varepsilon$; otherwise $u_\varepsilon \geq b$ and we choose $t = 0$. For $\varepsilon = 0$, if $h(x) < b$ every $t \in [h(x), L]$ is minimax optimal, and we tie-break by choosing $t = L$; otherwise $h(x) \geq b$ and we choose $t = 0$. □

Proposition A.6 (Robust Optimization Gap for Ski-Rental)**.** *For the contextual ski-rental problem, $\Gamma(\varepsilon) \leq 2\varepsilon$. The maximum pointwise loss is $M = b$ over the positive-radius family $\{\pi_\varepsilon : \varepsilon > 0\}$, and at most $\max\{b,\ L-b\} \leq L$ over the full family $\{\pi_\varepsilon : \varepsilon \geq 0\}$.*

Proof. From the proof of Proposition A.5, $\mathrm{Val}(\varepsilon, \ell_0) = \min(\ell_0+\varepsilon,\ b)$. Since $\mathrm{Opt}(\ell) = \min(\ell, b)$ and $\ell \mapsto \min(\ell, b)$ is 1-Lipschitz, the gap is

$$\mathrm{Val}(\varepsilon,\ell_0) - \mathrm{Opt}(\ell) = \min(\ell_0+\varepsilon, b) - \min(\ell, b) \leq |(\ell_0+\varepsilon) - \ell| \leq 2\varepsilon,$$

using $\ell \in B(\ell_0, \varepsilon)$. Equality holds at $\ell = \ell_0 - \varepsilon$ when $\ell_0 + \varepsilon < b$, so the bound is tight for small ε.

For M on the positive-radius family: for $\varepsilon > 0$ the policy either chooses $t = 0$, in which case $\mathrm{Cost}(0,\ell) - \mathrm{Opt}(\ell) = b - \min(\ell, b) \leq b$, or chooses $0 < t = h(x)+\varepsilon < b$, in which case the pointwise loss is 0 when $\ell \leq t$, is $t + b - \ell < b$ when $t < \ell < b$, and is $t < b$ when $\ell \geq b$. So $M = b$. For the full family, the additional case $\varepsilon = 0$ with $t = L$ (when $h(x) < b$) has pointwise loss $\ell - \min(\ell, b) = \max(\ell - b, 0) \leq L - b$, so the maximum loss over the full family is at most $\max\{b,\ L-b\} \leq L$. □

Two upper bounds on $\mathrm{Loss}(\pi_{\varepsilon^*})$ are available. First, applying Corollary 5.4 with $C = 2$, $\alpha = 1$, $M = b$ gives the standard $O(\eta^{2/3})$ rate. Second, at $\varepsilon = 0$ the tie-break $t = L$ when $h(x) < b$ keeps the pointwise loss bounded by $|h(x) - \ell|$, so $\mathrm{Loss}(\pi_0) \leq \mathbb{E}[|h(x) - \ell|] \leq \eta$. Since ε^* minimizes the loss over $\varepsilon \geq 0$, it inherits the better of these two.

Corollary A.7 (Ski-Rental Loss Bound)**.** *For the contextual ski-rental problem,*

$$\mathrm{Loss}(\pi_{\varepsilon^*}) \leq \min\Big\{\eta,\ 2^{5/3}\, b^{1/3}\, \eta^{2/3}\Big\},$$

and combined with the finite pseudo-dimension established in Proposition A.11 below, the ERM selector of Theorem 6.3 gives, with probability at least $1-\delta$,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq \min\Big\{\eta,\ 2^{5/3}\, b^{1/3}\, \eta^{2/3}\Big\} + O\left(L\sqrt{\frac{\log n + \log(1/\delta)}{n}}\right).$$

A.3 Contextual House Flipping

Recall the contextual house flipping problem from Section 3.3.

Proposition A.8 (Robust Policy for House Flipping)**.** *For the contextual house flipping problem, the robust policy is*

$$\pi_\varepsilon(x) = (h_v(x)+\varepsilon)\cdot \mathbf{1}\{h_{v'}(x) - h_v(x) > 2\varepsilon\}.$$

Proof. Since Cost $= -$revenue, minimizing worst-case cost is equivalent to maximizing worst-case revenue. Write $u_v := \min\{h_v(x)+\varepsilon, V\}$ and $\ell_{v'} := \max\{h_{v'}(x) - \varepsilon, 0\}$ for the upper end of the feasible v-range and the lower end of the feasible v'-range, after intersecting the metric ball with $[0,V]^2$. We solve $\max_{p\in[0,V]} \min_{(v,v')\in B(h(x),\varepsilon)\cap[0,V]^2} (v' - p)\cdot \mathbf{1}\{p \geq v\}$.

If $p < u_v$: the adversary can choose $v = u_v > p$, so $\mathbf{1}\{p \geq v\} = 0$ and worst-case revenue is at most 0.

If $p \geq u_v$: every feasible v satisfies $v \leq u_v \leq p$, so $\mathbf{1}\{p \geq v\} = 1$ and revenue is $v' - p$. The adversary minimizes revenue by choosing $v' = \ell_{v'}$, giving worst-case revenue $\ell_{v'} - p$. This is decreasing in p, so it is maximized at $p = u_v$, yielding revenue $\ell_{v'} - u_v$.

Hence the optimal worst-case revenue is $\max\{\ell_{v'} - u_v, 0\}$, attained either at the no-buy option (revenue 0) or at $p = u_v$ (revenue $\ell_{v'} - u_v$ when positive). We claim that $\ell_{v'} > u_v$ if and only if $h_{v'}(x) - h_v(x) > 2\varepsilon$. Indeed, if either clipping is active then $\ell_{v'} - u_v \leq 0$: the case $u_v = V$ gives $\ell_{v'} \leq V = u_v$, and the case $\ell_{v'} = 0$ gives $0 = \ell_{v'} \leq u_v$. So $\ell_{v'} > u_v$ requires both clippings inactive, and then $\ell_{v'} - u_v = h_{v'}(x) - h_v(x) - 2\varepsilon$. Conversely, $h_{v'}(x) - h_v(x) > 2\varepsilon$ together with $h_v(x) \geq 0$ gives $h_{v'}(x) > 2\varepsilon$, and together with $h_{v'}(x) \leq V$ gives $h_v(x) < V - 2\varepsilon$, so neither clipping bites and the same identity holds. The robust policy thus posts $p = h_v(x) + \varepsilon$ when $h_{v'}(x) - h_v(x) > 2\varepsilon$ and otherwise does not buy. □

Proposition A.9 (Robust Optimization Gap for House Flipping)**.** *For the contextual house flipping problem, $\Gamma(\varepsilon) \leq 4\varepsilon$ and $M \leq V$.*

Proof. From the proof of Proposition A.8, $\mathrm{Val}(\varepsilon, (v_0, v_0')) = -\max(v_0' - v_0 - 2\varepsilon, 0)$. Since $\mathrm{Opt}((v, v')) = -\max(v' - v, 0)$, the gap is $\max(v' - v, 0) - \max(v_0' - v_0 - 2\varepsilon, 0)$. Let $u = v' - v$ and $u_0 = v_0' - v_0$. Since $v' \leq v_0' + \varepsilon$ and $v \geq v_0 - \varepsilon$, we have $u \leq u_0 + 2\varepsilon$, so by the 1-Lipschitz property of $\max(\cdot, 0)$:

$$\max(u, 0) - \max(u_0 - 2\varepsilon, 0) \leq \max(u_0 + 2\varepsilon, 0) - \max(u_0 - 2\varepsilon, 0) \leq 4\varepsilon.$$

Equality holds at $v = v_0 - \varepsilon$, $v' = v_0' + \varepsilon$ when $v_0' - v_0 > 2\varepsilon$, so the bound is tight for small ε.

For M: when $p < v$ (no sale), the loss is $\max(v' - v, 0) \leq V$. When $p \geq v$ (sale occurs), the loss is $\max(v' - v, 0) - (v' - p) \leq p - v + \max(v - v', 0) \leq V$. So $M \leq V$. □

Corollary A.10 (House Flipping Loss Bound)**.** *Applying Corollary 5.4 with $C = 4$, $\alpha = 1$, and $M = V$ gives*

$$\mathrm{Loss}(\pi_{\varepsilon^*}) \leq 2^{7/3}\, V^{1/3}\, \eta^{2/3}.$$

Combined with the finite pseudo-dimension established in Proposition A.11 below, the ERM selector of Theorem 6.3 gives, with probability at least $1 - \delta$,

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq 2^{7/3}\, V^{1/3}\, \eta^{2/3} + O\left(V \sqrt{\frac{\log n + \log(1/\delta)}{n}}\right).$$

A.4 Sample Complexity of ERM Selection

The application corollaries above invoke the ERM selector of Theorem 6.3. Its guarantee depends on the pseudo-dimension of the one-parameter loss class

$$\mathcal{F} := \{\ell_\varepsilon : \varepsilon \geq 0\}, \qquad \ell_\varepsilon(x, \theta) := \mathrm{Cost}(\pi_\varepsilon(x), \theta) - \mathrm{Opt}(\theta).$$

We now show that this class has constant pseudo-dimension for each of the three applications. The proof uses only the piecewise-affine structure of the closed-form robust policies.

Proposition A.11. *The function class $\mathcal{F} = \{\ell_\varepsilon : \varepsilon \geq 0\}$ has pseudo-dimension at most 2 for pricing and house flipping, and at most 3 for ski-rental.*

Proof. We first record the loss functions induced by the closed-form policies from Appendix A.

- *Pricing.* Let

  $$p_\varepsilon(x) := \max\{h(x) - \varepsilon, 0\}.$$

  Then

  $$\ell_\varepsilon(x, v) = v - p_\varepsilon(x)\, \mathbf{1}\{v \geq p_\varepsilon(x)\}.$$

- *Ski-rental.* For $\varepsilon > 0$, let

$$t_\varepsilon(x) := (h(x) + \varepsilon)\,\mathbf{1}\{h(x) + \varepsilon < b\}.$$

Then

$$\ell_\varepsilon(x, \ell) = (t_\varepsilon(x) + b)\,\mathbf{1}\{t_\varepsilon(x) < \ell\} + \ell\,\mathbf{1}\{t_\varepsilon(x) \geq \ell\} - \min\{\ell, b\}.$$

At $\varepsilon = 0$, the tie-breaking convention in Proposition A.5 gives

$$t_0(x) = L\,\mathbf{1}\{h(x) < b\}.$$

- *House flipping.* Let

$$\Delta_h(x) := h_{v'}(x) - h_v(x), \qquad p_\varepsilon(x) := (h_v(x) + \varepsilon)\,\mathbf{1}\{\Delta_h(x) > 2\varepsilon\}.$$

Then

$$\ell_\varepsilon(x, (v, v')) = \max\{v' - v, 0\} - (v' - p_\varepsilon(x))\,\mathbf{1}\{p_\varepsilon(x) \geq v\}.$$

We use the following elementary counting observation. Fix points $z_1, \ldots, z_m$ in the sample domain and thresholds $r_1, \ldots, r_m \in \mathbb{R}$. For each i, define the binary label

$$B_i(\varepsilon) := \mathbf{1}\{\ell_\varepsilon(z_i) \geq r_i\}.$$

If each B_i changes value at most s times as ε ranges over an interval I, then the joint label vector

$$(B_1(\varepsilon), \ldots, B_m(\varepsilon))$$

can change value at most sm times on I. Hence at most $sm + 1$ distinct joint label patterns are realizable on I.

We now bound s for each application.

Pricing. For fixed (x, v), the function $\varepsilon \mapsto \ell_\varepsilon(x, v)$ is constant at v on $[0, h(x) - v]$, jumps down to an affine piece on $(h(x) - v, h(x))$ that rises from 0 to v, and is constant at v again on $[h(x), \infty)$, with breakpoints

$$\varepsilon = h(x) - v \qquad \text{and} \qquad \varepsilon = h(x),$$

when these values lie in $[0, \infty)$. Because the two constant pieces have the same value v and the affine piece is monotone, for any threshold r the label $\mathbf{1}\{\ell_\varepsilon(x, v) \geq r\}$ changes value at most twice.

House flipping. For fixed (x, v, v'), the only possible positive-radius breakpoints are

$$\varepsilon = v - h_v(x) \qquad \text{and} \qquad \varepsilon = \frac{h_{v'}(x) - h_v(x)}{2},$$

again only when these values are nonnegative. Between consecutive breakpoints, $p_\varepsilon(x)$ is either zero or affine in ε, and hence $\ell_\varepsilon(x, (v, v'))$ is either constant or affine in ε. More specifically, when $v > 0$, the two no-sale regions have the same loss value $\max\{v' - v, 0\}$, and the sale region, if nonempty, is affine; when $v < h_v(x)$, the left no-sale region is empty and the structure reduces to affine-then-constant. When $v = 0$, the function is affine up to the margin breakpoint and then constant at zero. In all cases, for any threshold r, the label $\mathbf{1}\{\ell_\varepsilon(x, (v, v')) \geq r\}$ changes value at most twice.

Thus, for pricing and house flipping, any m points with thresholds induce at most $2m + 1$ joint label patterns as ε ranges over $[0, \infty)$. If m points were shattered, all 2^m binary patterns would have to be realizable, so

$$2^m \leq 2m + 1.$$

This inequality fails for $m \geq 3$. Hence the pseudo-dimension is at most 2 for pricing and house flipping.

Ski-rental. For $\varepsilon > 0$, the possible breakpoints for fixed (x, ℓ) are

$$\varepsilon = \ell - h(x) \qquad \text{and} \qquad \varepsilon = b - h(x),$$

when these values are positive. On the intervals between these breakpoints, $\ell_\varepsilon(x, \ell)$ is either constant or affine, with at most one affine piece. The affine piece can cross any fixed threshold at most once, and each structural breakpoint can change the label at most once. Therefore $\mathbf{1}\{\ell_\varepsilon(x, \ell) \geq r\}$ changes value at most three times on $(0, \infty)$.

Consequently, over positive radii, m ski-rental samples induce at most $3m + 1$ joint label patterns. The isolated policy at $\varepsilon = 0$ may add one additional joint pattern, so the full family over $\varepsilon \geq 0$ realizes at most $3m + 2$ joint patterns. Shattering m points would require

$$2^m \leq 3m + 2.$$

This inequality fails for $m \geq 4$. Hence the pseudo-dimension is at most 3 for ski-rental. □

Plugging this constant pseudo-dimension bound into Theorem 6.3 gives the following application-level ERM guarantee.

Corollary A.12. *For each of the three applications, with probability at least $1 - \delta$,*

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \leq \mathrm{Loss}(\pi_{\varepsilon^*}) + O\left(M\sqrt{\frac{\log n + \log(1/\delta)}{n}}\right).$$

Here M is the uniform pointwise loss bound for the same robust-policy family over which ERM is performed.

The piecewise-affine structure used in Proposition A.11 also yields an efficient algorithm for computing the empirical ERM radius. Finite pseudo-dimension gives statistical efficiency through Theorem 6.3, but by itself does not imply computational efficiency.

Proposition A.13 (Computational efficiency of ERM)**.** *For each of the three applications, given n samples $(x_i, \theta_i)_{i=1}^n$, the empirical loss*

$$\widehat{L}(\varepsilon) := \frac{1}{n}\sum_{i=1}^{n} \ell_\varepsilon(x_i, \theta_i)$$

admits an empirical risk minimizer over $\varepsilon \geq 0$, or, when the infimum is achieved only as a one-sided limit, an approximate minimizer with arbitrarily small optimization error, that can be computed in $O(n \log n)$ time.

Proof. Each sample contributes at most two positive-radius breakpoints, so $\widehat{L}$ is piecewise affine on $(0, \infty)$ with at most $2n$ breakpoints. The relevant breakpoints are:

- *Pricing:*
$$\varepsilon = h(x_i) - v_i, \qquad \varepsilon = h(x_i).$$
- *Ski-rental:*
$$\varepsilon = \ell_i - h(x_i), \qquad \varepsilon = b - h(x_i).$$
- *House flipping:*
$$\varepsilon = v_i - h_v(x_i), \qquad \varepsilon = \frac{h_{v'}(x_i) - h_v(x_i)}{2}.$$

Only nonnegative breakpoints need to be retained. Sorting them and sweeping from left to right gives an $O(n \log n)$ algorithm, dominated by the sorting step. During the sweep, one tracks the current value and slope of $\widehat{L}$ on each open interval between consecutive breakpoints. Because the loss may jump at a breakpoint, the algorithm compares the exact empirical loss at each breakpoint with the one-sided values on the adjacent intervals. For ski-rental, the isolated value at $\varepsilon = 0$ is also evaluated separately. If the empirical infimum over positive radii is achieved only as a one-sided limit, the algorithm returns an arbitrarily close positive-radius approximate minimizer in the same time bound. □

The uniform-convergence guarantee in Theorem 6.3 is unchanged if the exact ERM is replaced by an approximate ERM whose optimization error is of smaller order than the statistical error.

B Structural Conditions for the Robust Gap

We now give two structural conditions under which the robust optimization gap $\Gamma(\varepsilon)$ is well controlled.

The first condition, *pessimistic representatives* (Appendix B.1), allows discontinuous cost functions. It captures problems in which every local uncertainty set contains a single worst-case state whose clairvoyantly optimal action is safe throughout the set. This condition explains, in a unified way, why all three applications in Appendix A have $\Gamma(\varepsilon) = O(\varepsilon)$ despite their action-dependent discontinuities.

The second condition, *Hölder continuity* (Appendix B.2), covers the smoother regime in which the cost varies continuously with the state. In this case $\Gamma(\varepsilon)$ also vanishes with ε, but robustification is not needed to obtain η-sensitivity: the follow-the-prediction policy itself already has vanishing loss.

B.1 Pessimistic Representatives

The applications in Appendix A all have discontinuous costs, but the discontinuities are structured. In each case, for every small uncertainty ball, there is a conservative state in the ball whose optimal action performs no worse at any other state in the ball. We formalize this property below.

Definition B.1 (Pessimistic representative)**.** *For a set $U \subseteq \Theta$, a state $q \in U$ is a* pessimistic representative *of U if there exists an action $a_q \in \mathcal{A}$ such that*

$$\mathrm{Cost}(a_q, q) = \mathrm{Opt}(q)$$

and

$$\mathrm{Cost}(a_q, \theta) \le \mathrm{Cost}(a_q, q) \qquad \textit{for all } \theta \in U.$$

We say that the problem admits pessimistic representatives for metric balls *if every metric ball $B(\theta_0, \varepsilon)$ admits a pessimistic representative.*

Thus a_q is clairvoyantly optimal at q, and q is pessimistic in the sense that using a_q incurs weakly lower cost at every other state in the uncertainty set. The robust policy can therefore hedge against the whole ball by acting as if the representative state q were realized.

Let

$$\omega_{\mathrm{Opt}}(r) := \sup_{\rho(\theta,\theta') \le r} |\mathrm{Opt}(\theta) - \mathrm{Opt}(\theta')|$$

denote the modulus of continuity of the clairvoyant value function.

Theorem B.2 (Robust gap under pessimistic representatives)**.** *Suppose the problem admits pessimistic representatives for metric balls. Then, for every $\varepsilon \ge 0$,*

$$\Gamma(\varepsilon) \le \omega_{\mathrm{Opt}}(2\varepsilon).$$

Consequently, if Opt *is uniformly continuous, then $\Gamma(\varepsilon) \to 0$ as $\varepsilon \to 0$. In particular, if* Opt *is (L, α)-Hölder continuous, then*

$$\Gamma(\varepsilon) \le L(2\varepsilon)^{\alpha}.$$

Proof. Fix $\theta_0 \in \Theta$ and let $U = B(\theta_0, \varepsilon)$. Let $q \in U$ be a pessimistic representative with associated action a_q.

For every action $a \in \mathcal{A}$,

$$\sup_{\theta \in U} \mathrm{Cost}(a, \theta) \ge \sup_{\theta \in U} \mathrm{Opt}(\theta),$$

because $\mathrm{Cost}(a, \theta) \ge \mathrm{Opt}(\theta)$ pointwise. Therefore

$$\mathrm{Val}(\varepsilon, \theta_0) \ge \sup_{\theta \in U} \mathrm{Opt}(\theta).$$

On the other hand, the representative action a_q gives

$$\mathrm{Val}(\varepsilon, \theta_0) \le \sup_{\theta \in U} \mathrm{Cost}(a_q, \theta) \le \mathrm{Cost}(a_q, q) = \mathrm{Opt}(q).$$

Since $q \in U$, we also have

$$\sup_{\theta \in U} \mathrm{Opt}(\theta) \ge \mathrm{Opt}(q).$$

Combining the three inequalities yields

$$\mathrm{Val}(\varepsilon, \theta_0) = \mathrm{Opt}(q) = \sup_{\theta \in U} \mathrm{Opt}(\theta).$$

Now fix any $\theta \in U$. Since both q and θ lie in $B(\theta_0, \varepsilon)$, the triangle inequality gives $\rho(q, \theta) \le 2\varepsilon$. Hence

$$\mathrm{Val}(\varepsilon, \theta_0) - \mathrm{Opt}(\theta) = \mathrm{Opt}(q) - \mathrm{Opt}(\theta) \le \omega_{\mathrm{Opt}}(\rho(q, \theta)) \le \omega_{\mathrm{Opt}}(2\varepsilon).$$

Taking the supremum over θ_0 and $\theta \in B(\theta_0, \varepsilon)$ proves the claim. □

The exact representative condition can be relaxed. Suppose that for every ball $U = B(\theta_0, \varepsilon)$ there exist $q \in U$ and $a_q \in \mathcal{A}$ such that

$$\sup_{\theta \in U} \mathrm{Cost}(a_q, \theta) \le \mathrm{Opt}(q) + r(\varepsilon),$$

where $r(\varepsilon) \to 0$ as $\varepsilon \to 0$. Then the same proof gives

$$\Gamma(\varepsilon) \le \omega_{\mathrm{Opt}}(2\varepsilon) + r(\varepsilon).$$

Thus exact pessimistic representatives are not essential; it is enough to have locally safe actions whose worst-case cost is near-optimal for some state in the ball.

Remark B.3. *The representative condition is stronger than continuity of* Opt. *Indeed, continuity of* Opt *alone is not enough to guarantee* $\Gamma(\varepsilon) \to 0$ *as* $\varepsilon \to 0$. *To see why, consider* $\Theta = [-1, 1]$, $\mathcal{A} = \{-1, +1\}$, *and*

$$\mathrm{Cost}(a, \theta) = \mathbf{1}\{a \neq \mathrm{sign}(\theta)\},$$

with the convention $\mathrm{sign}(0) = +1$. *Then* $\mathrm{Opt}(\theta) = 0$ *for all* θ, *so* Opt *is constant. However, for every* $\varepsilon > 0$, *the ball* $B(0, \varepsilon)$ *contains both a positive and a negative state. No single action is optimal on both sides:* $a = +1$ *incurs cost* 1 *at a negative state, while* $a = -1$ *incurs cost* 1 *at a positive state. Hence* $\mathrm{Val}(\varepsilon, 0) = 1$ *and* $\Gamma(\varepsilon) \ge 1$ *for every* $\varepsilon > 0$.

Corollary B.4 (The three applications admit pessimistic representatives)**.** *The contextual pricing, ski-rental, and house-flipping problems all admit pessimistic representatives for metric balls. Consequently,*

$$\Gamma(\varepsilon) \le 2\varepsilon \text{ for pricing, } \Gamma(\varepsilon) \le 2\varepsilon \text{ for ski-rental, and } \Gamma(\varepsilon) \le 4\varepsilon \text{ for house-flipping.}$$

These recover the robust-gap bounds in Propositions A.2, A.6 and A.9.

Proof. Pricing. Let

$$U = [v_0 - \varepsilon, v_0 + \varepsilon] \cap [0, 1],$$

and take

$$q = \underline{v} := \max\{v_0 - \varepsilon, 0\}.$$

Choose price $a_q = \underline{v}$. Every $v \in U$ satisfies $v \ge \underline{v}$, so the sale occurs at price $\underline{v}$. Therefore

$$\mathrm{Cost}(a_q, v) = -\underline{v} = \mathrm{Cost}(a_q, q) = \mathrm{Opt}(q) \qquad \text{for all } v \in U.$$

Thus q is a pessimistic representative. Since $\mathrm{Opt}(v) = -v$ is 1-Lipschitz, Theorem B.2 gives $\Gamma(\varepsilon) \le 2\varepsilon$.

Ski-rental. Let

$$U = [\ell_0 - \varepsilon, \ell_0 + \varepsilon] \cap [0, L],$$

and take

$$q = \bar{\ell} := \min\{\ell_0 + \varepsilon, L\}.$$

If $\bar{\ell} < b$, choose $a_q = \bar{\ell}$. Then every $\ell \in U$ satisfies $\ell \le \bar{\ell} = a_q$, so the decision-maker rents for the whole trip and

$$\mathrm{Cost}(a_q, \ell) = \ell \le \bar{\ell} = \mathrm{Cost}(a_q, q) = \mathrm{Opt}(q).$$

If $\bar{\ell} \ge b$, choose $a_q = 0$, i.e., buy immediately. Then

$$\mathrm{Cost}(a_q, \ell) \le b = \mathrm{Cost}(a_q, q) = \mathrm{Opt}(q) \qquad \text{for all } \ell \in U,$$

where the inequality only matters at $\ell = 0$. Hence q is a pessimistic representative in both cases. Since $\mathrm{Opt}(\ell) = \min\{\ell, b\}$ is 1-Lipschitz, Theorem B.2 gives $\Gamma(\varepsilon) \le 2\varepsilon$.

House-flipping. Let U be the ℓ_∞ ball around (v_0, v_0'), intersected with $[0, V]^2$. Define

$$\bar{v} := \min\{v_0 + \varepsilon, V\}, \qquad \underline{v}' := \max\{v_0' - \varepsilon, 0\},$$

and take

$$q = (\bar{v}, \underline{v}').$$

If $\underline{v}' > \bar{v}$, choose offer $a_q = \bar{v}$. Every $(v, v') \in U$ satisfies $v \le \bar{v} = a_q$ and $v' \ge \underline{v}'$, so the offer is accepted and

$$\mathrm{Cost}(a_q, (v, v')) = \bar{v} - v' \le \bar{v} - \underline{v}' = \mathrm{Cost}(a_q, q) = \mathrm{Opt}(q).$$

If $\underline{v}' \le \bar{v}$, choose $a_q = 0$. Then $\mathrm{Opt}(q) = 0$, and for every $(v, v') \in U$,

$$\mathrm{Cost}(a_q, (v, v')) = -v' \, \mathbf{1}\{v = 0\} \le 0 = \mathrm{Opt}(q).$$

Moreover, $\mathrm{Cost}(a_q, q) = 0 = \mathrm{Opt}(q)$, so q is again a pessimistic representative.

Finally,

$$\mathrm{Opt}(v, v') = -\max\{v' - v, 0\}$$

is 2-Lipschitz under the ℓ_∞ metric: the map $(v, v') \mapsto v' - v$ is 2-Lipschitz, and $z \mapsto \max\{z, 0\}$ is 1-Lipschitz. Therefore Theorem B.2 gives $\Gamma(\varepsilon) \le 4\varepsilon$. □

The pessimistic-representative theorem gives a structural explanation for the linear robust gaps derived explicitly in Appendix A. The closed-form robust policies in that section identify the corresponding safe action a_q for each problem.

B.2 Hölder Continuity

We now consider the smoother regime in which the cost function itself is regular in the state. Unlike the previous subsection, this condition rules out the threshold discontinuities that appear in our applications. In exchange, it implies not only that $\Gamma(\varepsilon)$ vanishes, but also that follow-the-prediction is already η-sensitive.

Definition B.5 (Hölder continuity)**.** *The cost function* Cost *is* (L, α)-Hölder continuous *in the state variable if there exist* $L \ge 0$ *and* $\alpha > 0$ *such that for all* $a \in \mathcal{A}$ *and all* $\theta, \theta' \in \Theta$*,*

$$|\,\mathrm{Cost}(a, \theta) - \mathrm{Cost}(a, \theta')| \le L\rho(\theta, \theta')^\alpha.$$

Proposition B.6 (Robust gap under Hölder costs)**.** *Suppose* Cost *is* (L, α)*-Hölder continuous in the state variable. Then, for every* $\varepsilon \ge 0$*,*

$$\Gamma(\varepsilon) \le 2L\varepsilon^\alpha.$$

Proof. First observe that Opt is also (L, α)-Hölder. Indeed, for any $\theta, \theta' \in \Theta$, let $a_{\theta'}$ be an optimal action at θ'. Then

$$\mathrm{Opt}(\theta) - \mathrm{Opt}(\theta') \le \mathrm{Cost}(a_{\theta'}, \theta) - \mathrm{Cost}(a_{\theta'}, \theta') \le L\rho(\theta, \theta')^\alpha.$$

Swapping θ and θ' gives the reverse inequality.

Now fix $\theta_0 \in \Theta$ and let $U = B(\theta_0, \varepsilon)$. Let a_0 be an optimal action at θ_0. Then

$$\mathrm{Val}(\varepsilon, \theta_0) \le \sup_{\theta \in U} \mathrm{Cost}(a_0, \theta) \le \mathrm{Cost}(a_0, \theta_0) + L\varepsilon^\alpha = \mathrm{Opt}(\theta_0) + L\varepsilon^\alpha.$$

For any $\theta \in U$, the Hölder continuity of Opt gives

$$\mathrm{Opt}(\theta) \ge \mathrm{Opt}(\theta_0) - L\varepsilon^\alpha.$$

Therefore

$$\mathrm{Val}(\varepsilon, \theta_0) - \mathrm{Opt}(\theta) \le 2L\varepsilon^\alpha.$$

Taking the supremum over θ_0 and $\theta \in B(\theta_0, \varepsilon)$ proves the result. □

Under the same condition, hedging is not needed to obtain vanishing loss: the follow-the-prediction policy already performs well.

Theorem B.7 (FtP under Hölder costs)**.** *Suppose* Cost *is* (L, α)*-Hölder continuous with* $\alpha \in (0, 1]$*. Then*

$$\mathrm{Loss}(\pi_{\mathrm{FtP}}) \le 2L\eta^{\alpha}.$$

In particular, π_{FtP} *is* η*-sensitive.*

Proof. Fix a realized pair (x, θ), and let

$$\widehat{a} = \pi_{\mathrm{FtP}}(x) \in \arg\min_{a \in \mathcal{A}} \mathrm{Cost}(a, h(x))$$

be the follow-the-prediction action. Let $a^* \in \arg\min_{a \in \mathcal{A}} \mathrm{Cost}(a, \theta)$ be a clairvoyant optimal action at the true state θ. By Hölder continuity,

$$\begin{aligned}
\mathrm{Cost}(\widehat{a}, \theta) &\le \mathrm{Cost}(\widehat{a}, h(x)) + L\rho(h(x), \theta)^{\alpha} \\
&\le \mathrm{Cost}(a^*, h(x)) + L\rho(h(x), \theta)^{\alpha} \\
&\le \mathrm{Cost}(a^*, \theta) + 2L\rho(h(x), \theta)^{\alpha} \\
&= \mathrm{Opt}(\theta) + 2L\rho(h(x), \theta)^{\alpha}.
\end{aligned}$$

Thus the pointwise excess loss is at most $2L\rho(h(x), \theta)^{\alpha}$. Taking expectations gives

$$\mathrm{Loss}(\pi_{\mathrm{FtP}}) \le 2L\,\mathbb{E}[\rho(h(x), \theta)^{\alpha}].$$

Since $\alpha \in (0, 1]$, the map $z \mapsto z^{\alpha/2}$ is concave on $\mathbb{R}_+$. Jensen's inequality therefore implies

$$\mathbb{E}[\rho(h(x), \theta)^{\alpha}] = \mathbb{E}[(\rho(h(x), \theta)^2)^{\alpha/2}] \le \left(\mathbb{E}[\rho(h(x), \theta)^2]\right)^{\alpha/2} = \eta^{\alpha}.$$

Combining the two inequalities proves the theorem. □

We restrict attention to $\alpha \in (0, 1]$, the usual nontrivial Hölder range. On length spaces, including intervals and convex subsets of Euclidean space with their standard metrics, an (L, α)-Hölder function with $\alpha > 1$ must be constant along paths, so this regime is typically degenerate. In particular, when the cost is Lipschitz in the state ($\alpha = 1$), follow-the-prediction achieves loss $O(\eta)$.

The applications in Appendix A lie outside this smooth regime. Their cost functions are discontinuous at action-dependent thresholds, and therefore are not Hölder continuous in the state for any $\alpha > 0$. The pessimistic-representative condition above explains why robustification can still achieve $\Gamma(\varepsilon) \to 0$ in these discontinuous settings.

B.3 Adverse Corners and Approximate Representatives

A useful specialization of Theorem B.2 applies to box-shaped state spaces with a fixed direction of adversity. Let $\Theta = \prod_{i=1}^{d} [\underline{\theta}_i, \overline{\theta}_i]$ carry the ℓ^∞ metric. Fix $s \in \{-1, +1\}^d$ and write $\theta \preceq_s \theta'$ when $s_i\theta_i \le s_i\theta'_i$ for every i. For example, increasing demand or decreasing a buyer's valuation can make a state more adverse.

Proposition B.8 (Monotone adverse corners)**.** *Suppose that* $\mathrm{Cost}(a, \theta) \le \mathrm{Cost}(a, \theta')$ *whenever* $\theta \preceq_s \theta'$*, for every* $a \in \mathcal{A}$*, and nominal optima are attained. The adverse corner* q *of* $B(\theta_0, \varepsilon)$*, given by*

$$q_i = \begin{cases} \min\{\theta_{0,i} + \varepsilon, \overline{\theta}_i\}, & s_i = +1, \\ \max\{\theta_{0,i} - \varepsilon, \underline{\theta}_i\}, & s_i = -1, \end{cases}$$

is a pessimistic representative. In particular, $\mathrm{Val}(\varepsilon, \theta_0) = \mathrm{Opt}(q)$ *and* $\Gamma(\varepsilon) \le \omega_{\mathrm{Opt}}(2\varepsilon)$*. The robust policy can be computed by solving the nominal problem at the adverse corner of the ball around* $h(x)$*.*

Proof. Every θ in the ball satisfies $\theta \preceq_s q$, and q belongs to the ball. Thus, for every action a,

$$\sup_{\theta \in B(\theta_0, \varepsilon)} \mathrm{Cost}(a, \theta) = \mathrm{Cost}(a, q).$$

Minimizing over a proves the equality of values and shows that any nominal optimizer at q is a pessimistic-representative action. The gap bound follows from Theorem B.2. □

The monotonicity requirement can also be relaxed. Suppose $\mathrm{Cost} = g+m$, where $g(a,\cdot)$ is uniformly L_g-Lipschitz and $m(a,\cdot)$ is nondecreasing under $\preceq_s$ for every action. If a_q is nominally optimal at the adverse corner, then, for every θ in the ball,

$$\mathrm{Cost}(a_q,\theta) \le g(a_q,q) + 2L_g\varepsilon + m(a_q,q) = \mathrm{Opt}(q) + 2L_g\varepsilon.$$

The approximate-representative argument above therefore gives

$$\Gamma(\varepsilon) \le \omega_{\mathrm{Opt}}(2\varepsilon) + 2L_g\varepsilon. \tag{1}$$

B.4 Coupled Capacity Planning

Consider a contextual multi-item inventory or capacity-planning problem, related to constrained newsvendor models [20]. Demands lie in $\Theta = \prod_{i=1}^d [0, D_i]$, with $D_i > 0$, and reserved capacities belong to a nonempty polytope $\mathcal{A} \subseteq \prod_{i=1}^d [0, D_i]$ that may encode coupled budget or network constraints. With nonnegative reservation costs c_i and shortage penalties p_i, let

$$\mathrm{Cost}(a,\theta) = c^\top a + \sum_{i=1}^d p_i(\theta_i - a_i)^+.$$

The cost is nondecreasing in each demand coordinate, so Proposition B.8 applies with $s_i = +1$. Given a prediction, set $q_i = \min\{h_i(x)+\varepsilon, D_i\}$. A robust action is obtained from the linear program

$$\min_{a,z}\ c^\top a + p^\top z \quad \text{subject to} \quad a \in \mathcal{A}, \qquad z_i \ge q_i - a_i, \quad z_i \ge 0 \quad (i = 1,\dots,d).$$

Thus robustification requires one nominal solve with modified demands and $O(d)$ work to form q.

For every a, the cost is $\|p\|_1$-Lipschitz in θ under ℓ^∞. Indeed, the positive-part map is 1-Lipschitz, so

$$|\,\mathrm{Cost}(a,\theta) - \mathrm{Cost}(a,\theta')| \le \sum_i p_i|\theta_i - \theta_i'| \le \|p\|_1\|\theta - \theta'\|_\infty.$$

Taking minima over the same feasible action set shows that Opt has the same Lipschitz constant. Consequently,

$$\Gamma(\varepsilon) \le 2\|p\|_1\varepsilon, \qquad M \le \sum_{i=1}^d (c_i + p_i)D_i < \infty,$$

where the second bound follows because all costs are nonnegative and at most the displayed sum. This example demonstrates that coupled, multidimensional actions pose no obstacle to the framework. Its costs are also Lipschitz, so Theorem B.7 already gives vanishing loss for FtP; robustness is not needed for that conclusion.

B.5 Inventory with a Fixed Stockout Fee

We next consider a single-period inventory model with proportional costs and a fixed stockout penalty, motivated by the inventory costs studied by Çetinkaya and Parlar [7]. Let $\Theta = \mathcal{A} = [0, D]$, with $D > 0$, use the absolute-value metric, and define

$$\mathrm{Cost}(a,\theta) = c_o(a-\theta)^+ + c_u(\theta - a)^+ + \kappa\mathbf{1}\{\theta > a\}, \qquad c_o, c_u, \kappa > 0.$$

The clairvoyant optimum is zero and uniquely chooses $a = \theta$. Costs are discontinuous at stockout and are not monotone in demand, because overage costs decrease as demand rises.

Proposition B.9 (Robust inventory policy and loss bound)**.** *Write $l = \max\{h(x) - \varepsilon, 0\}$ and $u = \min\{h(x)+\varepsilon, D\}$. For every $\varepsilon \ge 0$, a robust policy is*

$$\pi_\varepsilon(x) = \min\left\{u, \frac{c_o l + c_u u + \kappa}{c_o + c_u}\right\}. \tag{2}$$

Moreover, with $M_{\mathrm{inv}} := \max\{c_o D, c_u D + \kappa\}$,

$$\Gamma(\varepsilon) \le 2c_o\varepsilon, \qquad M \le M_{\mathrm{inv}}, \qquad \inf_{\varepsilon \ge 0} \mathrm{Loss}(\pi_\varepsilon) \le 2(2c_o)^{2/3} M_{\mathrm{inv}}^{1/3}\eta^{2/3}.$$

Proof. If $l = u$, matching that state is optimal. Otherwise actions outside $[l, u]$ are weakly improved by moving to the nearest endpoint. For $l \le a < u$, the worst-case cost over the interval is

$$\max\{c_o(a - l),\ c_u(u - a) + \kappa\}.$$

At $a = u$ the fee disappears, and the worst-case cost is $c_o(u - l)$. If $\kappa < c_o(u - l)$, the two affine terms intersect at $a = (c_o l + c_u u + \kappa)/(c_o + c_u) \in (l, u)$, with value strictly below $c_o(u - l)$; this action minimizes their maximum. If $\kappa \ge c_o(u - l)$, every $a < u$ has worst-case cost at least κ, so $a = u$ is optimal. These cases give Eq. (2).

The feasible action $a = u$ never stocks out within the ball and costs at most $c_o(u - l) \le 2c_o\varepsilon$. Since $\mathrm{Opt} \equiv 0$, this bounds Γ. On the whole action and state spaces, the overage case costs at most $c_o D$ and the shortage case at most $c_u D + \kappa$, proving the bound on M. Applying Corollary 5.4 with $C = 2c_o$ and $\alpha = 1$ proves the loss bound; when $\eta = 0$, radius zero gives zero loss directly. □

This example also gives a failure of FtP. Fix $h_0 \in (0, D)$, let $\mathcal{X} = \{x_0\}$, and let demand equal $h_0 - \eta$ or $h_0 + \eta$ with equal probability, where $0 < \eta \le \min\{h_0, D - h_0\}$. The predictor $h(x_0) = h_0$ is the conditional mean with RMSE η. Its unique nominal action orders h_0, and hence

$$\mathrm{Loss}(\pi_{\mathrm{FtP}}) = \tfrac{1}{2}(c_o + c_u)\eta + \tfrac{1}{2}\kappa \ge \kappa/2.$$

In contrast, when $2c_o\eta \le \kappa$, radius $\varepsilon = \eta$ orders $h_0 + \eta$ and incurs expected loss $c_o\eta$.

No nondegenerate interval has an exact pessimistic representative here: such a representative would require one action to cost zero throughout the interval, whereas zero cost requires $a = \theta$. The proportional-cost part is uniformly $\max\{c_o, c_u\}$-Lipschitz and the fixed-fee part is nondecreasing in demand. Thus Eq. (1) also applies, giving $\Gamma(\varepsilon) \le 2\max\{c_o, c_u\}\varepsilon$; the direct endpoint argument above gives the sharper coefficient $2c_o$.

Proposition B.10 (Efficient tuning for equal unit costs)**.** *Suppose $c_o = c_u = c$. For a fixed predictor, the inventory loss family has constant pseudo-dimension. Given n IID context–demand samples independent of predictor training, an exact ERM radius can be computed in $O(n \log n)$ arithmetic operations after evaluating the predictions. With probability at least $1 - \delta$, its loss satisfies*

$$\mathrm{Loss}(\pi_{\hat{\varepsilon}_{\mathrm{erm}}}) \le \inf_{\varepsilon \ge 0} \mathrm{Loss}(\pi_\varepsilon) + O\left((cD + \kappa)\sqrt{\frac{\log n + \log(1/\delta)}{n}}\right).$$

Proof. For a fixed prediction, l and u have at most two clipping breakpoints, and the width $u - l$ is nondecreasing. The action switches from u to $(l + u + \kappa/c)/2$ at most once. It is continuous and has at most four affine intervals. On each interval, crossing the realized demand creates at most one further breakpoint, so each sample loss has at most eight affine intervals, with separate values retained at their endpoints. Thresholding these pieces adds only a constant number of events per sample. For m sample–threshold pairs there are therefore $O(m)$ label patterns, including event values, which cannot realize all 2^m patterns for arbitrarily large m. The pseudo-dimension is constant, and Theorem 6.3 gives the statistical bound with $M = cD + \kappa$.

For computation, the action is constant for $\varepsilon \ge D$, so it suffices to search $[0, D]$. Each sample contributes $O(1)$ clipping, branch-switch, and demand-crossing events. Sort all $O(n)$ events, group coincident events, and sweep through them while maintaining the sum of the affine slopes and intercepts. At each event evaluate the exact equality values of the affected samples before updating to the next interval. This takes $O(n \log n)$ time. The action is continuous, and cost is lower semicontinuous in the action, including its downward jump at $a = \theta$. The empirical loss is therefore lower semicontinuous on $[0, D]$ and attains its minimum. Between events it is affine, and each endpoint value is no larger than its adjacent one-sided limits. Consequently some event, including 0 and D, attains the minimum, so this sweep returns an exact ERM radius. □

C Remaining Cases of Proposition 3.1

In each case below, the context space is trivial ($\mathcal{X} = \{x_0\}$), and the constructed predictor coincides with the conditional mean $\mathbb{E}[\theta \mid x_0]$, so it is minimum-variance with RMSE exactly η by construction.

Ski-rental. Assume $\eta \le b/2$, set $h(x_0) = b - \eta$, and draw $\ell \mid x_0 = h(x_0) + \eta\epsilon$ for ϵ an independent Rademacher random variable. At this prediction, $h(x_0) < b$, and the FtP cost-minimization $\arg\min_t \mathrm{Cost}(t, h(x_0))$ is degenerate: every $t \in [h(x_0), L]$ achieves the minimum cost $h(x_0)$. The natural point-response tie-break is $t = h(x_0) = b - \eta$ ("buy at the predicted trip end"), which is what we analyze below. The robust-policy tie-break $t = L$ in Proposition A.5 is the alternative choice; it does not exhibit the failure shown here, and indeed recovers the η rate in Corollary A.7.

With $t = b - \eta$: when $\epsilon = +1$, $\ell = b > t$, so the cost is $t + b = 2b - \eta$ and $\mathrm{Opt}(\ell) = b$, giving loss $b - \eta$; when $\epsilon = -1$, $\ell = b - 2\eta < t$, so the cost is $\ell = b - 2\eta = \mathrm{Opt}(\ell)$, giving loss 0. Hence $\mathrm{Loss}(\pi_{\mathrm{FtP}}) = \frac{1}{2}(b - \eta) \ge b/4$ for $\eta \le b/2$.

House flipping. Fix $C \in (0, V/2]$ and $\eta \le C$, and set $h_v(x_0) = \eta$ and $h_{v'}(x_0) = \eta + C$. Both lie in $[0, V]$, since $\eta \le C \le V/2$ gives $\eta + C \le V$. Draw $v \mid x_0 = h_v(x_0) + \eta\epsilon$ for ϵ an independent Rademacher random variable, and set $v' \mid x_0 = h_{v'}(x_0)$ deterministically. Then $v \in \{0,\, 2\eta\} \subset [0, V]$ and $v' = \eta + C \in [0, V]$, both within the state space. h_v and $h_{v'}$ are minimum-variance predictors, and the joint RMSE under the ℓ^∞ metric is η.

Since $h_{v'}(x_0) - h_v(x_0) = C > 0$, the FtP policy offers $p = h_v(x_0) = \eta$ (Proposition A.8 at $\varepsilon = 0$). When $\epsilon = +1$, $v = 2\eta > p$, so the offer is rejected; loss $= \max(v' - v, 0) = (\eta + C) - 2\eta = C - \eta$. When $\epsilon = -1$, $v = 0 \le p$, so the offer is accepted; loss $= \max(v' - v, 0) - (v' - p) = (\eta + C) - C = \eta$. Hence $\mathrm{Loss}(\pi_{\mathrm{FtP}}) = \frac{1}{2}(C - \eta) + \frac{1}{2}\eta = C/2$.

D Why Adaptation Is Necessary and How It Works

D.1 Proof of Proposition 3.2

The constructions below use a single context $\mathcal{X} = \{x_0\}$ and the same predictor for every positive error level η. A policy constructed from this predictor without access to the distribution, the error level, or calibration samples must therefore use the same action distribution for every η. We first compute the limiting loss of each fixed action, then extend the argument to randomized policies.

Proof of Proposition 3.2. Posted pricing. Fix $h(x_0) = 1/2$. For each $0 < \eta < 1/2$, let $\mathcal{D}_\eta$ assign equal probability to the values $v = 1/2 - \eta$ and $v = 1/2 + \eta$. The predictor is the conditional mean and has RMSE exactly η. The expected clairvoyant revenue is $\mathbb{E}[v] = 1/2$.

Let $\ell_\eta^{\mathrm{price}}(p)$ be the expected loss of posting a fixed price $p \in [0, 1]$. If $p < 1/2$, both values exceed p for all sufficiently small η, so the loss is $1/2 - p$. If $p = 1/2$, only the higher value buys, giving loss $1/4$ for every positive η. If $p > 1/2$, neither value buys for all sufficiently small η, giving loss $1/2$. Thus

$$\lim_{\eta \downarrow 0} \ell_\eta^{\mathrm{price}}(p) = g_{\mathrm{price}}(p) := \begin{cases} 1/2 - p, & p < 1/2, \\ 1/4, & p = 1/2, \\ 1/2, & p > 1/2. \end{cases}$$

Every value of g_{price} is strictly positive.

House flipping. Fix $h_v(x_0) = V/4$ and $h_{v'}(x_0) = V/2$. For each $0 < \eta < V/4$, let $v = V/4 - \eta$ or $V/4 + \eta$ with equal probability, and let $v' = V/2$ deterministically. Both predicted coordinates are their conditional means, and the joint RMSE under the ℓ^∞ metric is exactly η. Purchasing is profitable in both states, so the expected clairvoyant profit is $\mathbb{E}[v' - v] = V/4$.

Let $\ell_\eta^{\mathrm{flip}}(p)$ be the expected loss of making a fixed offer $p \in [0, V]$. If $p < V/4$, the offer is rejected in both states for all sufficiently small η, giving loss $V/4$. If $p = V/4$, only the lower current value accepts; the expected profit is $V/8$, giving loss $V/8$. If $p > V/4$, both states accept for all sufficiently small η, giving profit $V/2 - p$ and loss $p - V/4$. Hence

$$\lim_{\eta \downarrow 0} \ell_\eta^{\mathrm{flip}}(p) = g_{\mathrm{flip}}(p) := \begin{cases} V/4, & p < V/4, \\ V/8, & p = V/4, \\ p - V/4, & p > V/4. \end{cases}$$

Again, every value of g_{flip} is strictly positive.

Randomized policies. In either construction, let μ denote the fixed distribution of the policy's price or offer at x_0. Its randomization is independent of the unobserved state. The losses above lie in $[0, 1]$ for

pricing and $[0, V]$ for house flipping. Dominated convergence therefore gives, with g and ℓ_η denoting the corresponding functions,

$$\lim_{\eta \downarrow 0} \mathrm{Loss}_{\mathcal{D}_\eta}(\pi^h) = \lim_{\eta \downarrow 0} \int \ell_\eta(p)\, d\mu(p) = \int g(p)\, d\mu(p) > 0.$$

The final inequality holds because $g(p) > 0$ at every action. This positive limit may depend on the policy; a common positive constant for all policies is not needed. In particular, no fixed predictor-only policy can satisfy $\mathrm{Loss}_{\mathcal{D}_\eta}(\pi^h) \le f(\eta)$ throughout this family for a function f that tends to zero with η. □

This impossibility concerns a policy fixed across unknown error levels. It does not rule out policies that use the error level or calibration samples. It also does not assert failure at $\eta = 0$: with perfect predictions, following the prediction attains the clairvoyant optimum.

D.2 A Worked Pricing Example

We return to the pricing construction above, which is also the example in Proposition 3.1. Fix $0 < \eta < 1/2$. The robust price is $p(\varepsilon) = \max\{1/2 - \varepsilon, 0\}$, and its population loss is

$$L_\eta(\varepsilon) := \mathrm{Loss}_{\mathcal{D}_\eta}(\pi_\varepsilon) = \begin{cases} 1/4 + \varepsilon/2, & 0 \le \varepsilon < \eta, \\ \varepsilon, & \eta \le \varepsilon \le 1/2, \\ 1/2, & \varepsilon > 1/2. \end{cases} \tag{3}$$

Indeed, below η only the higher value buys, whereas at and above η both values buy. A radius of at least $1/2$ sets the price to zero. Consequently,

$$\arg\min_{\varepsilon \ge 0} L_\eta(\varepsilon) = \begin{cases} \{\eta\}, & 0 < \eta < 1/4, \\ \{0, \eta\}, & \eta = 1/4, \\ \{0\}, & 1/4 < \eta < 1/2, \end{cases} \qquad \min_{\varepsilon \ge 0} L_\eta(\varepsilon) = \min\{\eta, 1/4\}.$$

Thus a suitable radius gives vanishing loss, but a poorly chosen positive radius can perform worse than FtP, whose loss is $1/4$.

Now take n independent calibration observations $v_1, \ldots, v_n$ from $\mathcal{D}_\eta$, and write

$$\hat{q}_n := \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{v_i = 1/2 + \eta\}.$$

Minimizing empirical loss is equivalent to maximizing empirical revenue, because the empirical clairvoyant revenue is independent of the radius. The empirical revenue is

$$G_n(\varepsilon) := \frac{1}{n} \sum_{i=1}^{n} p(\varepsilon) \mathbf{1}\{v_i \ge p(\varepsilon)\} = \begin{cases} (1/2 - \varepsilon)\hat{q}_n, & 0 \le \varepsilon < \eta, \\ 1/2 - \varepsilon, & \eta \le \varepsilon \le 1/2, \\ 0, & \varepsilon > 1/2. \end{cases}$$

The best radius is therefore among $\{0, \eta\}$. Breaking a revenue tie in favor of zero gives the ERM rule

$$\hat{\varepsilon}_{\mathrm{erm}} = \begin{cases} 0, & \hat{q}_n \ge 1 - 2\eta, \\ \eta, & \hat{q}_n < 1 - 2\eta. \end{cases} \tag{4}$$

This calculation does not require prior knowledge of η: in this particular example, every observation reveals $\eta = |v_i - 1/2|$.

Since $n\hat{q}_n$ is binomial with parameters n and $1/2$, Hoeffding's inequality yields

$$\begin{aligned} \Pr[\hat{\varepsilon}_{\mathrm{erm}} \ne \eta] = \Pr[\hat{q}_n \ge 1 - 2\eta] &\le \exp\left(-8n(1/4 - \eta)^2\right), && 0 < \eta < 1/4, \\ \Pr[\hat{\varepsilon}_{\mathrm{erm}} \ne 0] = \Pr[\hat{q}_n < 1 - 2\eta] &\le \exp\left(-8n(\eta - 1/4)^2\right), && 1/4 < \eta < 1/2. \end{aligned}$$

For every fixed $\eta \ne 1/4$ in this range, ERM therefore selects the population-optimal radius with probability at least $1 - \exp(-8n(\eta - 1/4)^2)$. At $\eta = 1/4$, both candidate radii are population-optimal, so the selected policy is optimal with probability one for every sample size. At $\eta = 0$, the two values coincide, and radius zero has zero empirical and population loss.

E Implementing the Robust Policy - Contextual Linear Optimization

Appendix A derives the robust policy for each of the applications in closed-form, leading to simple implementations of the robust policy in each of these settings. Other settings may lead to significantly increased complexity for implementing the robust policy. Here, we discuss a wide class of settings where this is actually not the case: contextual linear optimization with convex constraints.

This setting was considered in [12]. We can describe this setting in our framework as follows: we take $\Theta = \mathbb{R}^d$ and $\mathcal{A} \subseteq \mathbb{R}^d$ a compact and convex set with $\mathrm{Cost}(a, \theta) = \langle \theta, a \rangle$. The metric ρ is induced by a norm $\|\cdot\|$. Given this setup, the robust problem given prediction $\widehat{\theta} = h(x)$ becomes

$$\pi_\varepsilon(x) = \arg\min_{a \in \mathcal{A}} \max_{\theta : \|\theta - \widehat{\theta}\| \leq \varepsilon} \langle \theta, a \rangle.$$

This problem can be rewritten as a single convex optimization problem:

$$\begin{array}{ll} \min\limits_{a} & \langle \widehat{\theta}, a \rangle + \varepsilon \|a\|_* \\ \text{s.t.} & a \in \mathcal{A}. \end{array}$$

Here $\|a\|_*$ is the dual norm corresponding to $\|\cdot\|$ defined as $\|a\|_* = \sup_x \{\langle a, x \rangle : \|x\| \leq 1\}$. Refer to [6] for additional details and the derivation of this reformulation. Note that the dual norm is in fact a norm, and is hence convex

This shows that implementing the robust policy for this setting amounts to solving a convex optimization problem involving the dual norm of $\|\cdot\|$. In the case of the ℓ_1 or ℓ_∞ norm (which are duals of each other), the convex objective above can be rewritten as a linear objective involving $O(d)$ additional variables and linear constraints, while in the case of the ℓ_2 norm (which is self-dual), the objective can be passed into the constraints as a single additional second-order cone constraint. If $\mathcal{A}$ is given by a polyhedron, then each of these cases reduce to either a linear program (LP) or a second-order cone program (SOCP), both of which are solvable in polynomial time [2, 10].